



Machine learning and the cross-section of cryptocurrency returns

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ABSTRACT

We employ a repertoire of machine learning models to investigate the cross-sectional return predictability in cryptocurrency markets. While all methods generate substantial economic gains—unlike in other asset classes—the benefits from model complexity are limited. Return predictability derives mainly from a handful of simple characteristics, such as market price, past alpha, illiquidity, and momentum. Contrary to the stock market, abnormal returns in cryptocurrencies originate from the long leg of the trade and persist over time. Furthermore, despite high portfolio turnover, most machine learning strategies remain profitable after trading costs. However, alphas are concentrated in hard-to-trade assets and critically depend on harvesting extreme returns on small, illiquid, and volatile coins.

1. Introduction

Cryptocurrency literature has documented a growing list of characteristics that predict cross-sectional returns. Some of them—such as momentum, size, or reversal (Bianchi et al., 2022; Liu et al., 2022)—are parallels of similar phenomena in equity markets; others, such as network activity measures (Cong et al., 2022; Liu & Tsyvinski, 2021), are inherently specific to cryptocurrencies. The emerging cryptocurrency “factor zoo”—comprising potentially noisy and correlated predictors—may require methods beyond simple portfolio sorts or cross-sectional regressions. Recent advances in machine learning methods appear to be a natural response to this task. Due to their capacity to handle vast multidimensional data, select best predictors, and account for nonlinearities and interactions (Giglio et al., 2022; Gu et al., 2020), machine learning models are well poised to face the cryptocurrency landscape.

In this paper, we combine machine learning with asset pricing research in order to gain new insights into cross-sectional return predictability in cryptocurrency markets. Building on a comprehensive dataset that covers >500 major coins and tokens listed across 250

exchanges over the years of 2017 to 2023, we identify, classify, and reproduce 40 cryptocurrency characteristics. Next, we use these to feed eight popular machine learning models; this includes dimension reduction techniques, raw and regularized regressions, tree methods, and neural networks. Finally, we explore their prediction performance. The aim of these exercises is not only to assess the machine learning effectiveness in the cryptocurrency world but—first and foremost—to enrich our understanding of pricing mechanisms within this new and growing asset class.

Our results could be summarized as follows. First, we demonstrate that machine learning techniques can be successfully applied to the cross-section of cryptocurrency returns. All our prediction models generate substantial economic gains. Admittedly, the prediction errors are relatively high, leading to seemingly disappointing predictive R^2 coefficients. Nevertheless—even though the machine learning models struggle to deliver impressive of prediction performance—they still work well enough to help separate winners from losers.

Interestingly, unlike in the stock market, the benefits from model complexity in cryptocurrencies are limited. The simplest methods prove the most effective. In particular, often-overlooked ordinary least squares

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visibly outperform regression trees and neural networks. This phenomenon places cryptocurrency markets in stark contrast to equities, where complex models typically dominate (Kelly et al., 2024). This observation can be interpreted in at least three ways. First, extracting profits from nonlinearities and interactions proves more difficult—or less useful—in the dynamically changing cryptocurrency universe. Second, it may signal that the cryptocurrency landscape may be less dimensional than equities. As a result, simpler models—such as that of Liu et al. (2022)—suffice to capture the cross-section of returns. Third, given the volatile cryptocurrency market so densely populated with outliers and extreme returns, existing data may be insufficient to train sophisticated models properly and extract valuable information. Simply speaking, more data and longer trading history may be necessary to capture complex patterns in cryptocurrency returns.

All our machine learning models can be forged into successful investment strategies via portfolio sorts. For the value-weighted portfolios, the highest risk-adjusted return is earned by the forecast combination, which equally weights outputs from individual models. In a nutshell, while all models have their benefits, merging them works best. A long-short combination strategy yields a mean weekly return of 2.37% at an annualized Sharpe ratio of 1.66. The abnormal returns survive controlling for common factor exposures, and the associated three-factor alpha amounts to 2.42%. Importantly, because space constraints prevent us from displaying the full details of our machine learning strategies, we supplement the paper with a visual tool to view them on a dedicated website.¹ It allows comparing the performance of all strategies discussed in this study.

Second, we uncover notable differences between the return predictability in cryptocurrencies and traditional asset classes. Evidence from the stock market documents that the profitability of anomaly-based strategies originates from the short leg of the trade (e.g., Avramov et al., 2013; Hong et al., 2000; Stambaugh et al., 2012) and gradually attenuates over time (e.g., McLean & Pontiff, 2016). In contrast, the profitability of cryptocurrency signals aggregated using machine learning routines is attributable almost solely to their long leg. The cross-sectional return pattern is L-shaped rather than linear, with the top quintile earning impressive alphas and the rest exhibiting mediocre performance. This implies that the short-selling constraints are not a critical factor shaping the return predictability of crypto assets.

Moreover, unlike in stocks, machine learning gains in cryptocurrency markets do not visibly decline over time. For example, the alpha of the combination forecast strategy in the second half of our study periods falls only slightly below that of earlier years. In other words, neither investor learning from finance literature nor market efficiency and liquidity improvements have yet advanced enough to undermine the effectiveness of machine learning methods. Noteworthy, this observation bears promising implications for market practitioners: harvesting cryptocurrency alphas from anomaly-based signals may still be possible.

Third, our findings highlight the main drivers of the cross-sectional variation in cryptocurrency returns. The machine learning models allow for pinpointing the essential return predictors among the “factor zoo”. Our variable importance analysis suggests that cryptocurrency returns are determined by a handful of uncomplicated signals—including share price, past alpha, momentum, and illiquidity. Notably, all these variables originate from simple data types—such as prices and returns. Moreover, when predictors are grouped into general economic categories, what matters the most are past returns and liquidity.

Our fourth contribution concerns the sources of cross-sectional return predictability in cryptocurrency markets. A deeper look at machine learning forecasts provides insights into the nature of asset pricing mechanisms. Our findings align most closely with the mispricing

explanation of return predictability: market inefficiencies emerge due to inefficiencies that cannot be easily arbitrated away. Consistent with this, machine learning alphas concentrate mainly in difficult-to-trade cryptocurrencies. While our strategies are profitable everywhere, they thrive particularly in market segments that are characterized by excessive limits to arbitrage: high idiosyncratic risk, illiquidity, and bid-ask spread. In other words, the return predictability exists to the extent that impediments to arbitrage prevent traders from fully eliminating inefficiencies. Overall, the alphas in the tercile of assets with the highest limits to arbitrage are roughly three to four times as high as in the tercile of easiest-to-trade cryptocurrencies.

Our fifth—and final—contribution pertains to an investor’s practical perspective. The cryptocurrency machine learning strategies produce phenomenal alphas, which can hardly be encountered in other asset classes. Their Sharpe ratios are impressive, though not as high as those of best machine learning strategies for developed or emerging stock markets (e.g., Gu et al., 2020; Liu et al., 2022). Can these profits be harvested in practice? We find that despite a sizeable portfolio turnover, most machine learning strategies survive net of transaction costs—even under restrictive assumptions. Admittedly, the trading costs can consume, on average, more than half of the payoffs of long-short machine learning strategies. Nonetheless, the remaining profits are still sizeable for most of the models.

However, cryptocurrency machine learning strategies come with four caveats. First, they exhibit substantial downside risk. Investors need to be prepared to lose even as much as 70–80% of invested capital. Second, they require taking massive positions in small and illiquid cryptocurrencies. In particular, the last point may be challenging or even unfeasible. Third, machine learning signals are relatively short-lived. In consequence, extending the forecast horizon and reducing the rebalancing frequency to just two weeks typically reduces the profitability by close to 30%. Fourth, portfolio profitability critically depends on harvesting extreme returns on volatile cryptocurrencies. Hence, potential profits decline considerably if their role is diminished in portfolio construction, e.g., by limiting their weight or imposing restrictions on the smallest and most volatile assets. Successful implementation of machine learning routines in cryptocurrency markets depends on sorting out these issues.

Our study connects with three main strains of finance research. First, we extend the discussion on cross-sectional return predictability in cryptocurrency markets. A growing list of studies has demonstrated dozens of signals that capture the cross-sectional variation in cryptocurrency returns. Examples include—but are not limited to—momentum, size, liquidity, reversal, downside risk, crypto-specific network and on-chain measures, as well as macroeconomic exposures (e.g., Liu et al., 2020; Liu & Tsyvinski, 2021; Zhang et al., 2021; Borri et al., 2022; Cong et al., 2022; Bianchi et al., 2022). Against this backdrop, our paper is most closely linked to studies that attempt to aggregate the multidimensionality of returns into a uniform pricing model. For example, Liu et al. (2022) propose a sparse three-factor model, Cong et al. (2022) extend its value and network adoption factors, and Bianchi and Babiak (2022) implement the instrumented principal component analysis of Kelly et al. (2019). Notably, in contrast to these studies, our unique machine learning framework enables us to gain insights into asset pricing that are not available from traditional models. Furthermore, it allows for the aggregation of a broad spectrum of cryptocurrency characteristics within a single model, providing a new perspective on return predictability at a comprehensive level.

Second, our paper adds to the research on machine learning applications to return predictability within financial markets—particularly in the cross-sectional context. Earlier studies concentrated on diverse asset classes, including U.S. equities (Freyberger et al., 2020; Gu et al., 2020, 2021; Avramov et al., 2023; Coqueret, 2022), international stocks (Leippold et al., 2022; Drobetz & Otto, 2021; Tobek & Hronec, 2021; Cakici et al., 2023; Azevedo et al., 2023; Fieberg et al., 2023; Hanauer & Kalsbach, 2023), corporate bonds (Bali et al., 2022), U.S. Treasury bonds

¹ https://azaremba.shinyapps.io/Cryptocurrency_machine_learning/.

(Bianchi et al., 2021), and commodities (Rad et al., 2023; Struck & Cheng, 2020). Cryptocurrencies have thus remained largely uncharted territory.

Notably, a relatively rich literature has applied machine learning to predict time-series asset returns (Bekiros et al., 2016; Cakici et al., 2024; Gupta et al., 2018; Neely et al., 2014; Zhou et al., 2023). This area of literature is also very active in cryptocurrencies, where researchers try to predict the time-series behavior of either bitcoin (Balcilar et al., 2017; Jaquart et al., 2021; Demir et al., 2018; Huang et al., 2019) or several different coins (Akyildirim et al., 2021; Fang et al., 2021; Wang et al., 2022). In contrast, the literature on cross-sectional predictions is limited to a few papers. The few sole exceptions include Liu et al. (2023), who mainly focus on predicting daily data with the eXtreme Gradient Boosting algorithm, Alessandretti et al. (2018), who study 1681 coins over a three-year period with gradient-boosted regression trees and long short-term memory networks, and Jaquart et al. (2021), who focus on predicting their outperformance or underperformance of the market median return within a sample of the 100 largest coins. Given this context, and in the spirit of Gu et al. (2020), we focus directly on cross-sectional return predictability and the related implications for both understanding asset pricing and practical trading.

Third, our findings relate to the evidence on economic constraints on return predictability. Existing literature focuses mainly on equities. For example, Hong et al. (2000), Fama and French (2008, 2012), and Hou et al. (2020) show that anomalies concentrate primarily in small and illiquid securities. Pontiff (2006) and McLean (2010) accentuate the role of idiosyncratic volatility limiting arbitrage. Stambaugh et al. (2012, 2015) and Avramov et al. (2013) argue that most abnormal returns come from short legs of anomaly portfolios. Novy-Marx and Velikov (2016, 2019), Chen and Velikov (2023) document the sensitivity of profits from cross-sectional return predictions to trading costs. Finally, abundant literature—encompassing the works of Chordia et al. (2014), McLean and Pontiff (2016), Calluzzo et al. (2019), and Linnainmaa and Roberts (2018)—demonstrate the gradual decrease in return predictability over time.² Against this background, our article is particularly associated with Avramov et al. (2022)—who explore economic restrictions on machine learning implementation effectiveness in equity markets. Similarly, they conclude that machine learning methods extract profitability mainly from difficult-to-arbitrage securities.

The remainder of the study proceeds as follows. Section 2 discusses our data and findings. Section 3 presents the baseline empirical findings on the prediction performance of machine learning models, as well as the contribution of individual crypto characteristics. Section 4 concentrates on machine learning portfolios. Section 5 scrutinizes the role of limits to arbitrage. Section 6 concerns a practical investor perspective. Finally, Section 7 concludes the study.

2. Data and methods

This section summarizes the data and methods used in this study. We begin by presenting our dataset and the sample of cryptocurrency characteristics. We then describe the machine learning models employed, as well as the methods of evaluating their performance.

2.1. Data sources and sample preparation

Following Bianchi and Babiak (2022), we combine data from several databases to maximize market coverage and availability of cryptocurrency characteristics. Specifically, we source OHLC price and volume data from [CryptoCompare.com](https://cryptocompare.com), market capitalizations from [CoinMarketCap](https://coinmarketcap.com), and blockchain activity from [IntoTheBlock.com](https://intotheblock.com). Our price and volume data provider, CryptoCompare, aggregates survivorship

bias-free transaction data from >250 centralized exchanges around the world to provide an accurate price estimation. The market data is volume-weighted across the exchanges, linking the data source's importance with trading activity. As a result, smaller trading venues are given less emphasis in the aggregation process than the big and liquid markets. CryptoCompare has been recommended by Alexander and Dakos (2020) due to its superior reliability; it is frequently employed as a prime data source (Bianchi et al., 2022; Borri, 2019; Borri & Shakhnov, 2022; Lucchini et al., 2020). On the other hand, IntoTheBlock—used by Cong et al. (2022), Bianchi and Babiak (2022), and Hoang and Baur (2022), among others—is one of the most common sources of on-chain activity.

Cryptocurrency tickers used by various providers may differ. Hence, in order to ensure correct matching coins from CryptoCompare and IntoTheBlock, we use a two-step procedure. In the first pass, we pair the cryptocurrencies by their full name and the ticker. However, cryptocurrency names may also exhibit minor discrepancies across databases. Hence, in the second pass, we manually verify all other cryptocurrencies on a case-by-case basis. Cryptocurrencies that cannot be matched are entirely removed from the sample. In consequence, our procedures value the quality of our sample over the sheer quantity of coins covered.

We clean our sample with a series of filters to eliminate potential data errors, mainly following Bianchi and Babiak (2022). We begin with static screens. We exclude (i) stablecoins, e.g., USDT or DAI (either centralized or algorithmically stabilized); (ii) coins backed by or tracking prices of precious metals, such as gold; (iii) cryptocurrencies serving as collaterals for derivative platforms, e.g., SNX; and (iv) so-called “wrapped coins,” including Wrapped Bitcoin (WBTC). The cryptocurrency classification for these filters is based on [CoinMarketCap.com](https://coinmarketcap.com).

Next, we also apply a battery of dynamic filters. First, following Cong et al. (2022), we delete any observations with non-positive trading volume, market capitalization, or price. Second, as in Bianchi et al. (2022), we mitigate issues with “fake” or “erroneous” volume by dropping observations with the ratio of daily traded volume to market capitalization exceeding one. Third, to align our study with practice, we exclude the smallest cryptocurrencies—which may be difficult to trade. To be precise, we adopt the approach of Liu et al. (2022): setting a minimum threshold of \$1 million for cryptocurrency market capitalization. Finally, similarly to Bianchi and Babiak (2022), we discard the extreme returns. Specifically, we winsorize all returns at −99% and 300% levels. This operation aims to mitigate the impact of potential database errors, as well as limit the dependence of our results on outliers.

After we apply all the filters, our sample contains 552 distinct cryptocurrencies. The study period runs from July 1, 2017, to January 26, 2023, and the start time is defined as 00:00:00 UTC. Naturally, the number of coins is not constant over time and grows gradually along with the evolution of the cryptocurrency markets. Fig. 1 provides a snapshot of the filtered research sample. As seen in Panel A, the number of available assets increased from 30 in July 2017 to reach 250 in late 2018. Notably, the cross-section may seem modest when compared to the existing cryptocurrency universe; at the time of writing, [CoinMarketCap.com](https://coinmarketcap.com) reports >24 thousand coins traded at different exchanges. Nevertheless, by focusing on larger and more tradeable tokens, we simultaneously capture most of the global market capitalization. As seen in Panel C, the total market value of the sample is below less than \$100 billion in 2017 to exceed two trillion in 2021—effectively reflecting most of the tradeable cryptocurrency universe.

Finally, Panel B of Fig. 1 illustrates the cross-sectional composition of our research sample. An average token in our selection has 871 trading days' worth of data.³ A total of 124 coins have a history longer than

² Notably, several studies—including Jacobs (2016) and Jacobs and Müller (2020)—contest the conclusions on declining return predictability.

³ Unlike stocks, cryptocurrencies are traded 365 days a year. As a result, 774 trading days imply roughly 29 months.

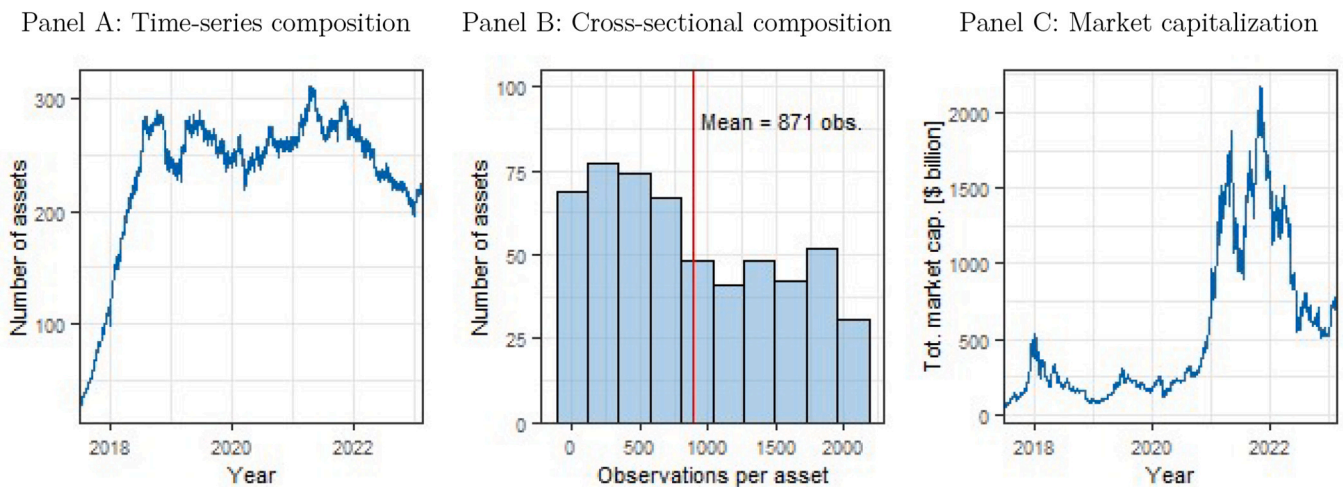


Fig. 1. A Snapshot of the Research Sample.

The figure provides an overview of the sample of cryptocurrencies used in this study. Panel A presents the time-series evolution of the number of cryptocurrencies included in the sample. Panel B illustrates the cross-sectional distribution of the number of daily observations available per cryptocurrency. The red vertical line indicates the average number of observations across all assets in the sample. Panel C displays the sample's aggregate market capitalization (in USD billion) over time. The sample comprises 552 unique cryptocurrencies; the study period runs from July 1, 2017, to January 26, 2023. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

1500 trading days. On the other hand, the sample also includes a substantial representation of cryptocurrencies' relatively short history, and 152 coins have less than one year of data available.

2.2. Cryptocurrency characteristics

Our machine-learning forecasting models require inputs in the form of coin characteristics. Hence, we identify, classify, and reproduce 40 return-predicting variables from finance literature. We build our selection mainly on Liu et al. (2022) and Bianchi and Babiak (2022), which we complement with more recent findings from cryptocurrency research. We also use these source papers to follow possibly close the calculation procedure. Below, we briefly summarize the predictors, while Table 1 contains a detailed description.

We group the signals into six broad categories based on their underlying economic intuition. The first group—the measures of *on-chain activity*—encompasses new addresses and their dynamics (*new_add*, *new_add_ch*), active addresses (*active_add*), zero-balance addresses (*zero_bal*), addresses-with-balance growth (*ba_growth*), network-to-market ratio (*bm*), new address-to-price ratio (*ap*), and active addresses to network value (*aanv*). These signals are sometimes regarded as proxies for cryptocurrency value (Pagnotta & Buraschi, 2018; Cong et al., 2022; Liebi, 2022).

Next, we consider an array of variables associated with *liquidity* that has been explored in cryptocurrency literature (Brauneis et al., 2021; Dong et al., 2022; Han, 2023; Li et al., 2021; Liu et al., 2022; Zhang & Li, 2023): trading volume (*volume*), market value (*size*), bid-ask spread (*bidask*), illiquidity ratio (*illiq*), turnover (*turn*), and detrended turnover (*dto*). As in Leirvik (2021), Bianchi and Babiak (2022), and Dong et al. (2022), we also account for measures of variability in liquidity—namely, turnover and trading volume volatility (*std_dto*, *std_vol*), standardized abnormal turnover (*sat*) of Garfinkel et al. (2023), and 30- and 60-day volume shocks (*volsh_30d*, *volsh_60d*).

The third category includes various proxies for security-level price volatility and risk that have been found to either positively or negatively correlate with future cryptocurrency payoffs (Burggraf & Rudolf, 2021; Dobrynska & Dubrovskiy, 2022; Jia et al., 2021; Zhang et al., 2021; Zhang & Li, 2020): realized volatility (*rvol*), the capital asset pricing model (CAPM), beta (*beta*), idiosyncratic risk (*ivol*), and value-at-risk (*var*).

The fourth group, associated with *past returns*, is relatively broad. It encompasses assorted variables that are derived from past price changes. These signals capture the positive or negative correlation between past and future coin performance (Bianchi et al., 2022; Grobys & Sapkota, 2019; Jia et al., 2022; Liu et al., 2020; Liu & Tsyvinski, 2021; Shen et al., 2020; Tzouvanas et al., 2020). Specifically, we calculate several momentum measures motivated by Liu et al. (2022): seven-day, 13-day, 23-day, 31-day, and intermediate momentum (*r7_2*, *r13_2*, *r22_2*, *r31_2*, *r30_14*); two reversal signals: daily reversal (*r2_1*) and long-term reversal (*r360_31*), as well as two other measures: the 90-day high (*90dh*) and the CAPM alpha (*alpha*).

Besides the average past returns, we also study their *distribution*. In addition to skewness (*skew*) and kurtosis (*kurt*), this category also includes the maximum and minimum daily returns (*max*, *min*)—examined initially by Bali et al. (2011). Numerous works (e.g., Grobys & Junttila, 2021; Jia et al., 2021; Lin et al., 2021; Liu et al., 2021; Ozdamar et al., 2021) suggest that such distributional properties may contain information about future coin returns.

Finally, our set also includes several variables classified as *other*—which do not fit well into the categories above. These include the nominal price (*prc*), analyzed by Miller and Scholes (1982); the cross-sectional seasonality (*seas*) from Keloharju et al. (2016); and the variables associated with distribution perception distortions: the salience theory measure (*st*) by Cosemans and Frehen (2021) and the chronological return ordering (*cro*) variables of Mohrschladt (2021). The cryptocurrency counterparts of these anomalies have been scrutinized by Zaremba et al. (2020), Cai and Zhao (2024), Chen et al. (2022), and Liu et al. (2022)—among others.

Table 2 presents the summary statistics for the main variables in the sample. We report the averages, standard deviations, skewness coefficients, and key percentiles. Notably, most variables have asymmetric distributions. Many of them—such as on-chain measures, size, or volume—exhibit positive skewness. This phenomenon reflects the specific market structure, with a small number of very large cryptocurrencies (such as Bitcoin or Ethereum). On the other hand, multiple liquidity variables display negative skewness—suggesting a long tail of illiquid coins.

Table 1
Cryptocurrency Characteristics.

No.	Characteristic	Symbol	Definition
<i>On-chain activity</i>			
(1)	New addresses	new_add	The logged number of new addresses per coin, where a new address refers to a unique address appearing for the first time in a transaction of the native coin in the network (Liu et al., 2022).
(2)	New addresses change	new_add_ch	A weekly change in the logged number of all new addresses per coin over the last week versus the week before, where a new address refers to a unique address appearing for the first time in a transaction of the native coin in the network (Liu et al., 2022).
(3)	Active addresses	active_add	The logged number of unique addresses active in the network, either as a sender or receiver (Pagnotta & Buraschi, 2018). The calculations are limited to the addresses that were active in successful transactions.
(4)	Zero-balance addresses	zero_bal	The logged number of zero-balance addresses (Borri et al., 2022)
(5)	Addresses-with-balance growth	ba_growth	The weekly change in the logged value of addresses with balance (Cong et al., 2022).
(6)	Network-to-market value	bm	As in Pagnotta and Buraschi (2018), the network-to-market value is calculated as a logged ratio of the cumulative number of unique addresses to the total market value (see Market value for the calculation details).
(7)	New address-to-price ratio	ap	A logged ratio of new addresses per coin over the last week to the coin market price (Liu et al., 2022).
(8)	Active addresses to network value	aanv	The average number of active addresses over the past 30 days to the total market value (Liebi, 2022).
<i>Liquidity</i>			
(9)	Trading volume	volume	The logged total dollar value of all native tokens transferred across wallets - both across and within centralized exchanges.
(10)	Market value	size	The logged market capitalization calculated by multiplying the total supply of coins or tokens by the current market price of each coin or token. (Liu et al., 2022)
(11)	Bid-ask spread	bidask	An estimation of the bid-spread calculated based on 30 days of OHLC data as a simple average of two approximations by Corwin and Schultz (2012) and Abdi and Ranaldo (2017).
(12)	Illiquidity ratio	illiq	The price impact measure of Amihud (2002) calculated as the average 90-day ratio of the absolute value of daily returns over the daily trading volume measured in USD.
(13)	Turnover	turn	The last day's dollar trading volume (see Trading volume) over the market value (see Market Value) (Datar et al., 1998).
(14)	De-trended turnover	dto	As in Bianchi and Babiak (2022), the de-trended turnover is calculated in two steps. First, we compute the daily excess turnover as the daily turnover (see Turnover) minus the value-weighted average daily market turnover. Second, we detrend the obtained value by its 180-day median.

Table 1 (continued)

No.	Characteristic	Symbol	Definition
(15)	Turnover volatility	std_dto	Residuals' standard deviation from a regression of daily turnover on a constant using a 30-day estimation period.
(16)	Trading volume volatility	std_vol	The logged value of residuals' standard deviation from a regression of daily trading volume on a constant using a 30-day estimation period.
(17)	Standardized abnormal turnover	sat	Following Garfinkel et al. (2023), standardized abnormal turnover is calculated in two steps. First, we calculate the change in turnover as the difference between the last day's turnover (see Turnover for definition) and its average over the prior 30 days. Second, we divide the outcome by the 30-day standard deviation of daily turnover values.
(18)	Volume shock (30 days)	volsh_30d	Log-deviation of trading volume from its rolling 30-day average, as in Llorente et al. (2002) and Babiak et al. (2022).
(19)	Volume shock (60 days)	volsh_60d	Log-deviation of trading volume from its rolling 60-day average, as in Llorente et al. (2002) and Babiak et al. (2022).
<i>Risk</i>			
(20)	Realized volatility	rvol	Daily realized volatility calculated based on 30 days of OHLC prices using the estimator of Yang and Zhang (2000).
(21)	CAPM beta	beta	The market beta from the Capital Asset Pricing Model estimated using a trailing 30-day period. As in Lewellen and Nagel (2006), the beta is calculated as the sum of two slope coefficients from the regression of daily cryptocurrency returns on the contemporaneous and one-day-lagged market excess returns. The market portfolio return is the value-weighted average return of all cryptocurrencies in the sample.
(22)	Idiosyncratic volatility	ivol	The standard deviation of the residuals from the regression of daily excess cryptocurrency returns on the daily market portfolio excess returns estimated using a trailing 30-day period. The market portfolio return is the value-weighted average return of all cryptocurrencies in the sample.
(23)	Value-at-risk	var	The historical empirical value-at-risk computed as the 5th percentile of daily returns over a rolling 90-day period.
<i>Past returns</i>			
(24)	Daily reversal	r2_1	Return on the previous day.
(25)	7-day momentum	r7_2	Cumulative return from seven to two days before return prediction.
(26)	13-day momentum	r13_2	Cumulative return from 13 to two days before return prediction.
(27)	22-day momentum	r22_2	Cumulative return from 22 to two days before return prediction.
(28)	31-day momentum	r31_2	Cumulative return from 31 to two days before return prediction.
(29)	Intermediate momentum	r30_14	Cumulative return from 30 to 14 days before return prediction.
(30)	Long-term reversal	r360_31	Cumulative return from 360/31 days before return prediction (Cong et al., 2022).
(31)	Closeness to the 90-day high	90dh	Following the logic of George and Hwang (2004), the closeness to the 90-day high is the last day's price over the maximum price over the previous 90 days. The estimation

(continued on next page)

Table 1 (continued)

No.	Characteristic	Symbol	Definition
(32)	CAPM alpha	alpha	period follows Babiak and Bianchi (2022). An intercept from the regression of daily excess cryptocurrency returns on the daily market portfolio excess returns based on a trailing 30-day period. The market portfolio return is the value-weighted average return on all cryptocurrencies in the sample.
<i>Distribution</i>			
(33)	Skewness	skew	Skewness of the daily return distribution calculated over a rolling 90-day period.
(34)	Kurtosis	kurt	Kurtosis of the daily return distribution calculated over a rolling 90-day period.
(35)	Maximum daily return	max	The maximum daily return over the last 30 days.
(36)	Minimum daily return	min	The minimum daily return over the last 30 days.
<i>Other</i>			
(37)	Salience theory	st	The salience theory variable is calculated closely following the multistep procedure in Cosemans and Frehen (2021) using a rolling 30-day estimation period. We use market portfolio return as the reference rate and set the parameters $\theta = 0.1$ and $\delta = 0.7$.
(38)	Chronological return ordering	cro	As in Mohrschladt (2021), the chronological return ordering variable is calculated as the correlation between daily returns over the last 30 days and the corresponding number of trading days until the end of the rolling 30-day estimation window.
(39)	Seasonality	seas	Average same-weekday return calculated over a rolling 20-week period.
(40)	Price	prc	Logged cryptocurrency price at the end of the previous day.

The table presents the 40 cryptocurrency characteristics used as inputs to machine learning models. *No.* is the running number, and *Symbol* indicates the acronym used to denote the variable in the paper. The table spans two pages.

2.3. Machine learning models

We build on Gu et al. (2020) and use a general additive prediction model to capture the association between the cryptocurrency returns and their characteristics:

$$r_{i,t+1} = E_t(r_{i,t+1}) + \varepsilon_{i,t+1}, \quad (1)$$

with $r_{i,t+1}$ denoting the excess return on token $i = 1, \dots, N_T$ in the week (seven days) $t = 1, \dots, T$. Most cryptocurrency asset pricing studies typically focus on explaining daily (e.g., Bianchi et al., 2022; Bianchi & Babiak, 2022) or weekly (e.g., Cong & He, 2019; Liu et al., 2022; Liu & Tsyvinski, 2021) returns. We opt for the latter, a less granular, approach for two reasons. First, the weekly returns are less noisy—allowing us to focus more on the underlying return drivers.⁴ Second, the less frequent portfolio reconstruction is more aligned with an investor's perspective as it generates lower trading costs.

We compute the expected returns $E_t(r_{i,t+1})$ as a constant function of

cryptocurrency characteristics:

$$E_t(r_{i,t+1}) = g(z_{i,t}), \quad (2)$$

where $z_{i,t}$ denotes the vector of stock characteristics, which contains the 40 variables described in Section 2.1. The function $g(z_{i,t})$ estimates the rates of return that solely rely on information from token i and day t . In other words, it does not take into account the characteristics of other cryptocurrencies or from earlier periods. Moreover, its precise form is not specified. The approximated functions differ across the machine learning algorithms and may be parametric or nonparametric, linear or nonlinear.

As in Bianchi and Babiak (2022), we consistently predict returns and construct strategies that assume a skipping in portfolio formation equals one day. In other words, we forecast returns based on data from day $d-1$, use them to form portfolios in d , and track their returns over the next week (i.e., from day $d+1$ to $d+7$). This one-day lag not only aligns well with the practical investor perspective but also allows for the mitigation of potential microstructure issues associated with bid-ask bounce (Bouri et al., 2022; Zaremba et al., 2021).

We train all models to forecast future returns by minimizing the out-of-sample mean squared forecast error:

$$MSFE_{t+1} = \frac{1}{N_{t+1}} \sum_{i=1}^{N_{t+1}} (\hat{\varepsilon}_{i,t+1})^2, \quad (3)$$

with $\hat{\varepsilon}_{i,t+1}$ indicating the individual prediction error for the cryptocurrency i , and N_{t+1} representing the number of tokens at $t+1$. As a general rule, we aim to find a forecasting model with the best prediction accuracy among a pool of candidates.

Machine learning literature offers a broad—and growing—number of prediction models. Our selection of a sample of representative methods builds on several seminal studies from asset pricing literature; this includes Gu et al. (2020), Bali et al. (2022), and Leipold et al. (2022). In consequence, we consider eight distinct models—which are briefly summarized below. Regarding further implementation details, we closely reproduce procedures from Cakici et al. (2023).

Ordinary Least Squares Regression (OLS). The OLS regression assumes fitting a multiple predictive regression that utilizes all features as model inputs. The model is relatively simple and does not involve regularizing, hyperparameters tuning, or validation. Nonetheless, it is highly prone to overfitting—especially in a high-dimensional setting (Gu et al., 2020).

Partial Least Squares (PLS). PLS is an effective dimension-reduction technique that accounts for the association between covariates and security returns. Advocated by Kelly and Pruitt (2013, 2015), the three-pass PLS regression concentrates on the features most strongly correlated with cryptocurrency payoffs. Specifically, the individual predictors are aggregated into composite factors to maximize the correlation between them and future returns. Subsequently, these newly created factors are used as inputs to predictive regressions. The tuning parameter is the number of components in the regression.

Penalized Linear Regressions (LASSO, ENET). Penalize linear regressions to cope with the overfitting problem by imposing a penalty term on slope coefficients. We employ two popular regularization approaches: the least absolute shrinkage and selection operator (LASSO), and the elastic net (ENET) of Zou and Hastie (2005). LASSO penalizes the model proportionally to the absolute values of its coefficients. On the other hand, ENET's penalty function combines the component of LASSO and the ridge regression of Hoerl and Kennard (1970)—which concentrates on squared coefficients. In our case, both components are weighted equally. The primary benefit of ENET over LASSO is its effectiveness in dealing with a correlation between covariates (for details, see Zou & Hastie, 2005; Diebold & Shin, 2019). The tuning parameter in both models is the penalty term.

Tree Models (RF, GBRT). The tree models are flexible non-

⁴ Aggregating returns removes much of the noise observed at higher frequencies (Chesnay & Jondeau, 2001). This is mainly used in the volatility literature (Andersen et al., 2011), but also in the topology approach (Rudkin et al., 2023). The shorter the time frame, the more problematic it is to separate real market movements from noise (Black, 1986; Cont, 2010).

Table 2
Statistical Properties of Cryptocurrency Characteristics.

	Mean	St. dev.	Skew	Percentiles						
				1st	5th	25th	50th	75th	95th	99th
On-chain measures										
new_add [x10 ⁴]	11.48	113.10	14.59	0.00	0.00	0.00	0.00	0.01	0.91	218.80
new_add_ch [x10 ⁴]	−0.03	23.58	−249.58	−8.44	−0.03	0.00	0.00	0.00	0.02	8.23
active_add	10.65	2.20	0.93	2.08	8.40	9.69	10.41	11.23	14.10	19.52
zero_bal	8.72	3.39	−0.18	0.00	1.95	7.97	9.32	10.27	12.72	19.42
ba_growth	0.70	0.07	50.26	0.65	0.69	0.69	0.69	0.69	0.71	0.78
bm	−6.29	2.09	−1.68	−15.16	−9.33	−7.37	−6.11	−5.00	−3.36	−2.22
ap	−12.63	1.72	−0.40	−17.49	−15.66	−13.56	−12.47	−11.55	−10.18	−8.28
aanv	−13.20	1.81	−0.90	−18.93	−16.66	−14.07	−12.94	−12.06	−10.75	−9.36
Liquidity										
volume	10.97	3.46	−0.32	1.92	4.92	8.99	11.15	13.12	16.48	19.10
size	17.10	2.15	1.03	13.89	14.24	15.51	16.81	18.30	20.99	23.71
bidask (%)	4.29	14.51	21.66	0.00	0.00	0.49	2.09	4.52	12.95	32.21
illiq [x10 ²]	0.82	7.24	11.39	0.00	0.00	0.00	0.00	0.00	0.55	18.10
turn (%)	1.86	5.61	7.60	0.00	0.00	0.05	0.31	1.30	8.16	27.41
dto (%)	0.36	4.31	6.50	−7.48	−1.85	−0.30	−0.01	0.28	3.51	15.62
std_dto (%)	1.28	2.94	4.83	0.00	0.01	0.08	0.30	1.03	6.07	16.22
std_vol	11.28	2.80	0.23	5.22	6.82	9.39	11.20	13.05	16.09	18.53
sat	0.01	1.10	1.60	−1.78	−1.20	−0.65	−0.30	0.40	2.23	4.06
Risk										
rvol (%)	14.91	38.21	16.31	2.74	4.12	6.65	9.43	13.95	30.31	99.93
beta	0.98	0.77	−0.73	−1.08	−0.05	0.64	0.99	1.32	1.99	2.95
ivol (%)	7.88	5.98	2.04	1.12	2.08	3.92	6.07	9.82	20.02	30.06
volsh_30d	−0.59	1.56	−2.20	−6.91	−3.47	−1.02	−0.31	0.21	1.21	2.21
volsh_60d	−0.71	1.67	−1.96	−7.20	−3.82	−1.25	−0.41	0.21	1.26	2.27
var (%)	−0.13	0.08	−3.67	−0.47	−0.26	−0.15	−0.11	−0.09	−0.06	−0.04
Past returns										
r2_1 (%)	0.26	10.42	1.36	−25.57	−13.43	−4.12	−0.19	3.70	15.33	36.67
r7_2 (%)	0.50	21.05	2.57	−46.60	−26.65	−9.36	−0.88	7.84	31.84	71.42
r13_2 (%)	1.14	32.96	4.42	−60.66	−37.74	−14.95	−2.12	11.52	50.00	112.76
r22_2 (%)	2.37	47.88	5.66	−74.79	−48.79	−21.44	−4.16	15.42	73.29	172.91
r31_2 (%)	4.06	64.50	8.45	−83.41	−56.29	−26.74	−6.18	18.80	93.85	230.60
r30_14 (%)	1.80	41.41	5.27	−69.40	−44.13	−18.84	−3.21	13.85	63.30	145.76
r180_60 (%)	84.02	618.93	12.26	−100.00	−97.42	−80.32	−48.96	28.72	593.99	2267.50
90dh (%)	62.55	1841.55	689.46	11.86	21.92	41.06	59.21	77.81	98.07	112.05
alpha (%)	0.20	1.84	−0.16	−4.21	−1.80	−0.57	0.01	0.77	3.07	6.08
Distribution										
skew	0.60	1.17	0.17	−2.62	−1.03	−0.02	0.50	1.15	2.62	4.06
kurt	4.52	5.92	2.88	−0.77	−0.09	0.99	2.46	5.73	15.94	29.35
min (%)	−18.59	12.45	−2.32	−69.83	−42.30	−22.28	−15.37	−10.83	−6.38	−3.74
max (%)	24.16	18.76	1.66	3.41	6.05	11.32	18.09	30.35	66.28	91.16
Other										
st [x10 ²]	−0.69	0.32	−1.99	−1.91	−1.25	−0.83	−0.64	−0.48	−0.31	−0.19
cro	0.00	0.17	0.00	−0.41	−0.28	−0.11	0.00	0.12	0.28	0.41
seas (%)	−0.43	4.18	−7.67	−12.65	−4.78	−1.61	−0.24	1.08	3.83	8.01
prc	−2.39	3.32	0.56	−9.74	−7.16	−4.59	−2.62	−0.60	3.81	7.22

The table presents the basic statistical properties of the 40 cryptocurrency characteristics used as inputs to machine learning models. The explanation of variable symbols seen in the leftmost column is available in Table 1. The reported values are calculated using a pooled sample of all daily observations. The sample comprises 552 unique cryptocurrencies, and the study period runs from July 1, 2017, to January 26, 2023.

parametric algorithms that effectively capture returns’ interactions and nonlinearities. They divide observations (cryptocurrency characteristics) into distinct subcategories, typically called “leaves.” A tree is constructed in stages, and the splitting variables and decision nodes determine the tree’s structure. At each splitting point, a splitting variable generates two disjoint branches. The tree continues to grow until the terminal leaves are reached.

Individual trees tend to overfit the data, so they require heavy regularization. We choose two popular approaches: random forests (RF) and gradient-boosted regression trees (GBRT). RF relies on the bootstrap aggregation algorithm called “bagging” (Breiman, 2001). Specifically, it assumes averaging multiple trees based on bootstrapped subsamples on the original data. The GBRT technique, in turn, assumes fitting subsequent trees based on earlier trees’ residuals and forming ensemble predictions by aggregating numerous trees consecutively multiplied by the learning rate (0.1). We fit the GBRT mode using the least-squares boosting method (Breiman, 2001; Hastie et al., 2008) with between 100 and 200 learning cycles.

Feed-Forward Neural Networks (FFNN). The feed-forward neural

networks comprise (i) an “input layer,” i.e., cryptocurrency characteristics; (ii) a number of “hidden layers” containing activation and functions that transform the features; and (iii) an “output layer” that converts the hidden layers’ outcomes into return predictions. The information flows through the neurons, from the input through hidden to the output layer, and eventually aggregates into forecasts.

The neural networks allow for the accounting of non-linearities and interactions, and their flexibility increases with the number of layers. As in Bali et al. (2022), we use FFNN with one hidden layer with the idea of capturing the benefits of shallow learning, as in Gu et al. (2020). The hidden layer comprises eight neurons employing the rectified linear units as the activation function. We optimize the trees using the Adam algorithm of Kingma and Ba (2014).

Forecast Combination. To create an ensemble forecast, the forecast combination (COMB) method weighs a number of predictions from various models. Combining several predictions into one has strong statistical roots (Bates & Granger, 1969; Clemen, 1989; Timmermann, 2006). Merging the models helps diminish their variance, eventually reducing the forecast error (Petropoulos et al., 2022). As in Bali et al.

(2022), we obtain the COMB forecast by calculating a simple equal-weighted average of the predictions from seven individual models discussed above: OLS, PLS, LASSO, ENET, RF, GBRT, and FFNN.

We estimate the model's parameters, tune their hyperparameters, and assess their performance using standard methods from machine learning literature. We split the sample into three consecutive subperiods, keeping their temporal order: an extending training period of 175 days, a fixed validation period of 75 days, and the testing period comprising the subsequent week. Hence, the initial in-sample period equals 250 days and runs from July 1, 2017, to March 7, 2018, and the first testing period is the week of March 8 to 14, 2018. The training sample is used to estimate the model's parameters subject to pre-specified hyperparameters (specific to a given model type). Next, we use the subsequent validation period to optimize the hyperparameters to minimize the objective loss function. Notably, to minimize the information loss, we use overlapping weekly data at the daily frequency to train and validate the models. Last, we test the model's accuracy using the following week. The testing week is not included in the training or validation samples. We re-estimate the model each week and reproduce this procedure until we reach the end of the entire study period. We assume an extending training window, which means that the length of the training increases gradually week-by-week, while the validation, and testing samples are held constant and rolled forward at each reestimation.

2.4. Performance evaluation

Our baseline measure of forecasting accuracy is the out-of-sample predictive R^2 coefficient:

$$R_{OOS}^2 = 1 - \frac{\sum_{(i,t) \in T_3} (r_{i,t+1} - \hat{r}_{i,t+1})^2}{\sum_{(i,t) \in T_3} r_{i,t+1}^2}, \quad (4)$$

where $\hat{r}_{i,t+1}$ and $r_{i,t+1}$ indicated forecasted and realized weekly returns on the cryptocurrency i in week $t + 1$, and T_3 denotes the testing sample. As in Gu et al. (2020), we estimated R_{OOS}^2 using the entire sample of all weekly return observations pooled across cryptocurrencies and time. In addition, to offer more context, we supplement R_{OOS}^2 with the absolute mean squared error (MAE)—calculated using the same data as R_{OOS}^2 .

In addition to the above, following the arguments of Cakici et al. (2023), we supplement R_{OOS}^2 with the simple average cross-sectional Pearson correlation coefficient ($\bar{\rho}_p$). This measures allows to capture whether the predictions order cryptocurrencies in line with their ex-post performance. In other words, they provide an intuitive snapshot of the relationship between the model expectations and realized profits.

Finally, to conduct pairwise comparisons of different machine learning models, we employ the modified test of Diebold and Mariano (1995), abbreviated as DM. The DM test statistic compares mean squared forecast errors to evaluate which model yields more precise predictions. Our implementation closely adheres to Drobetz and Otto (2021), with the DM statistic computed as:

$$DM_{a,b} = \frac{\bar{d}_{a,b}}{\hat{\sigma}_{\bar{d}_{a,b}}}, \quad (5)$$

where $\bar{d}_{a,b,t+1} = MSFE_{t+1}^{(a)} - MSFE_{t+1}^{(b)}$ represents the differences in the monthly mean squared forecast errors of models a and b ; $\bar{d}_{a,b} = \frac{1}{T} \sum_{t=1}^T \bar{d}_{a,b,t+1}$ indicates the time-series average of these differences, and $\hat{\sigma}_{\bar{d}_{a,b}}$ is the Newey and West (1987) adjusted standard error. The DM test statistics follow the standard normal distribution, and we interpret them in two distinct ways. First, for individual pairwise comparisons, we establish a standalone 5% significance threshold corresponding to a $|t\text{-stat}|$ of 1.96. Secondly, since we simultaneously examine eight models,

we address the multiple hypothesis problem by applying the Bonferroni correction (for discussion, see, for example, Harvey et al., 2016). The adjusted threshold for the t -statistics is 2.73.⁵

3. Baseline empirical findings

This section summarizes the prediction performance of machine learning methods. First, we summarize the overall measures of forecasting accuracy. Next, we explore the contribution of individual cryptocurrency characteristics.

3.1. Prediction performance

Table 3 presents the prediction performance for different machine learning methods, revealing remarkable disparities compared to conventional asset classes. Most notably, the overall predictability seems rather disappointing. The R_{OOS}^2 are negative for all prediction models, ranging from -0.21% (OLS) to -5.89% (FFNN). This strikingly contrasts evidence from equity and bond markets (Gu et al., 2020; Leippold et al., 2022; Bali et al., 2022), where the R_{OOS}^2 are typically low but positive. The low R_{OOS}^2 scores are associated with substantial return variance in the cryptocurrency world. The second row of Table 3 presents the mean absolute forecast errors (MAE) for different models. The MAE values are remarkably high, typically reaching about 16%, resulting from abundant extreme returns on crypto assets.

Do the negative R_{OOS}^2 values and high prediction errors signify that machine learning models are ineffective in forecasting the cross-section of cryptocurrency returns? Not necessarily. As Leitch and Tanner (1991) propose, there might be only a tenuous connection between statistical metrics and economic profitability. The statistical measures mentioned above rely on level predictions, while the profitability of typical portfolios from univariate sorts is derived from rank order accuracy. Machine learning models—especially in the cryptocurrency universe—may generate extreme and fluctuating estimates (resulting in poor statistical metrics); however, they may still accurately predict the rank order, leading to relatively better economic profitability. In other words, while the variance of predicted and realized returns may blur their cross-sectional relationship in the R_{OOS}^2 metric, the link between the two may still exist.

The bottom row of Table 3 displays the average weekly cross-sectional correlation coefficients. Their overview uncovers one notable difference: the $\bar{\rho}$ measures consistently exceed zero for all models considered—spanning from 0.015 (FFNN) to 0.046 (GBRT). Notably, even though the correlation levels may seem very low, they could be sufficient for producing substantial economic gains. This is due to idiosyncrasies in a portfolio setting being canceled out, which boosts the resulting predictability. As a result, even the least sophisticated models—such as OLS—can still effectively distinguish future winners from losers.

Beyond the sheer level of return predictability, Table 3 reveals another interesting dissimilarity to the equity universe: the relative effectiveness of different methods. Machine learning studies of traditional asset classes typically highlight significant benefits from reducing overfitting, accounting for nonlinearities, and capturing interactions. However, the cryptocurrency evidence fails to confirm these patterns. The OLS model, often disregarded in the equity literature, produces the highest R_{OOS}^2 in our array - albeit still negative (-0.21%). Dimension reduction techniques and penalized regressions bring no improvement; the R_{OOS}^2 coefficient for PLS, LASSO, and ENET is even lower than for OLS. The tree methods work somewhat better, yielding higher $\bar{\rho}$

⁵ For other methods of evaluation and comparison of machine learning forecasting models, see, e.g., Wu et al. (2013), Fan et al. (2018), Deng et al. (2019), Zhu et al. (2022), Lv and Wang (2023).

Table 3
Prediction Performance of Machine Learning Models.

Panel A: Standalone performance measures								
	OLS	PLS	LASSO	ENET	RF	GBRT	FFNN	COMB
Predictive out-of-sample R^2 (R^2_{OOS})	−0.21	−1.61	−1.37	−1.37	−1.02	−1.10	−5.89	−0.91
Mean absolute forecast error (MAE)	15.84	15.91	15.98	15.98	15.99	16.02	16.71	15.95
Average Pearson correlation ($\bar{\rho}$)	0.027	0.019	0.019	0.019	0.047	0.046	0.015	0.039
Panel B: Pairwise model comparisons								
	PLS	LASSO	ENET	RF	GBRT	FFNN	COMB	
OLS	−2.02	−1.95	−1.95	−1.56	−1.62	<u>−3.67</u>	−1.47	
PLS		1.37	1.38	<u>3.06</u>	2.43	<u>−3.17</u>	3.03	
LASSO			1.20	<u>3.75</u>	2.07	<u>−3.35</u>	<u>2.94</u>	
ENET				<u>3.75</u>	2.06	<u>−3.35</u>	<u>2.94</u>	
RF					−0.77	<u>−3.52</u>	0.87	
GBRT						<u>−3.52</u>	1.24	
FFNN							3.80	

The table presents the prediction performance of the eight machine learning models used in this study: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage and selection operator (LASSO), elastic net (ENET), gradient boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). Panel A reports the baseline performance measures: the predictive out-of-sample R^2 coefficients (R^2_{OOS}) computed as in [Gu et al. \(2020\)](#), root-mean absolute forecast errors (MAE), as well as average monthly Pearson ($\bar{\rho}$). R^2_{OOS} and MAE are expressed in percentages. Panel B presents the pairwise comparisons of the machine learning models using modified [Diebold and Mariano \(1995\)](#) tests (DM). A positive (negative) DM value implies that the model indicated in the top row outperforms (underperforms) the model indicated in the left-most column. The bold font denotes the values significant at the 5% level in standalone pairwise comparisons ($|t\text{-stat}| > 1.96$), and the underline indicates the 5%-significance incorporating the Bonferroni adjustment for the multiple hypothesis framework ($|t\text{-stat}| > 2.73$). The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018.

measures, but their R^2_{OOS} still fall behind OLS. The performance of FFNN is even worse. Apparently, the most sophisticated models cannot adapt to the extremely volatile and dynamic cryptocurrency environment. Finally, [Table 3](#) also discloses the strong performance of the COMB method; this corroborates earlier observations by [Cakici et al. \(2023\)](#) and [Bali et al. \(2022\)](#) from international equity and US bond markets. Forecast combinations reduce model variance and thus prediction error. As a result, their R^2_{OOS} is −0.91%, beating all other methods expected for OLS. In addition, the average Pearson correlation is 0.039, surpassing that of OLS.

[Table 3](#), Panel B, reports the pairwise model comparisons using the modified test by [Diebold and Mariano \(2021\)](#). The outcomes of this analysis are often inconclusive, with many insignificant values. Nevertheless, the results align with the key patterns observed in Panel A. First, no model can significantly beat OLS. Second, the tree models work relatively well, significantly outperforming penalized regressions and dimension reduction techniques. On the other hand, FFNN substantially lags behind other models. Finally, COMB once again emerges as a leading contender. This model remains unbeaten by other models while consistently outperforming more than half of them. Clearly, individual models have strengths and weaknesses, but combining them effectively leverages their strong points.

Overall, our findings indicate that the classical gains from complex machine learning models may be smaller for cryptocurrencies than stocks. Machine learning models benefit from digesting vast amounts of data with multidimensional and nonlinear patterns. However, if the cross-sectional structure of cryptocurrencies is simpler—with fewer independent return drivers, mainly linear patterns, and fewer interdependencies between them—the power of machine learning may not be fully unleashed. Likewise, complex machine learning models profit from stable and detectable patterns in asset returns. Training and validating them may be challenging in a highly volatile and dynamically changing environment. Also, this factor may contribute to the over-performance of more straightforward and robust algorithms, such as OLS.

3.2. Which cryptocurrency characteristics matter?

One of the unique traits of machine learning methods is that they help to pinpoint which of the variables matter for future returns. Therefore, they allow introducing some order into the “factor zoo”. Though the cryptocurrency asset pricing literature is not as rich as for stocks, yet, the number of documented cross-sectional returns already reaches into the dozens. Hence, we are interested in the relative importance of these signals.

To measure the contribution of specific variables to overall return predictability, we compute variable importance (VI) by following the approach developed by [Kelly et al. \(2019\)](#). To be precise, for each predictor, we calculate in R^2_{OOS} resulting from setting all variables to zero while keeping all else fixed. This allows us to capture the critical determinants of the cross-sectional variation in cryptocurrency returns while simultaneously accounting for the full battery of 40 signals in the system.

3.2.1. Individual importance

[Fig. 2](#) illustrates the VI of the 40 characteristics considered in this study. First, most models largely agree on the critical drivers of cryptocurrency returns. The top five variables include market price (*prc*), CAPM alpha (*alpha*), skewness (*skew*), illiquidity (*illiq*), and 30-day momentum (*r31_2*). Moreover, PLS emphasizes the role of bid-ask spread (*bidask*). On the other hand, the tree models accentuate the value at risk (*var*), the distance to the 90-day high (*90dh*), and the last day’s return (*r2_1*). Nevertheless, besides these differences, the selection of key variables is relatively similar across models.

Interestingly, the crucial predictors in [Fig. 2](#) align with those from extant cryptocurrency literature. For example, the seminal paper of [Liu et al. \(2022\)](#) pinpoints two essential determinants of the cross-section of cryptocurrency returns: past performance (momentum) and size (market capitalization). Likewise, two measures of past performance—*r31_2* and *capmaa*—belong to the most commonly selected variables. Admittedly, the precise specification (sorting period, market-return adjustment) may matter, but they both capture essentially similar phenomenon. Furthermore, while most models do not select *size* among the top predictors, all of them choose its close cousin: *prc*. Notably, [Liu](#)

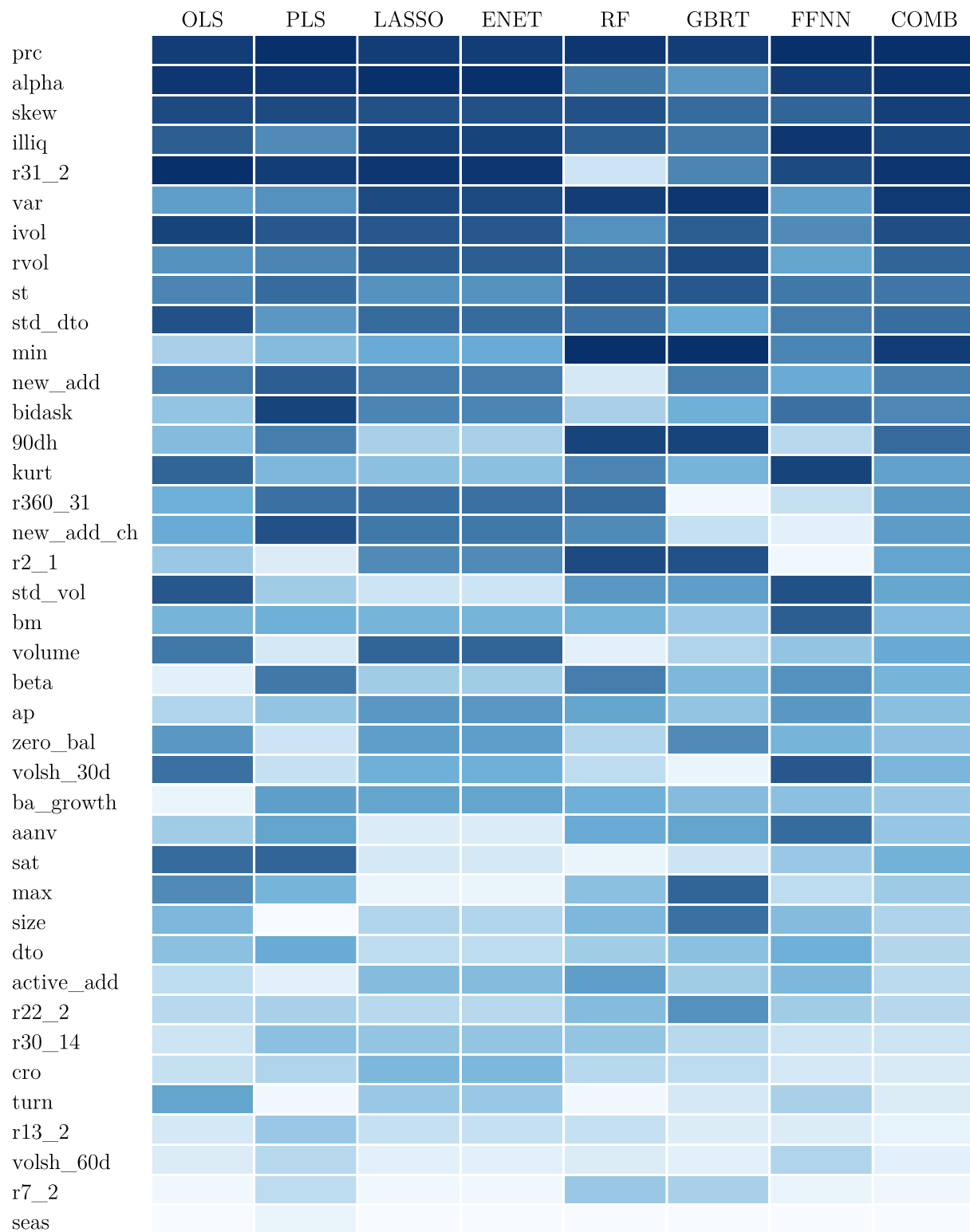


Fig. 2. Variable Importance per Characteristic.

The figure presents the rankings of 40 return predictors covered in the study in terms of their average total model contribution. The variable importance (VI) is computed as the reduction of the overall OOS R^2 resulting from excluding a given variable from the model. We consider eight machine learning models: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage and selection operator (LASSO), elastic net (ENET), gradient-boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). The colour gradients represent the VI rank; the dark blue (white) represents the most influential (least influential) characteristics. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

et al. (2022) classifies both variables into the same size category and find that *prc* yields higher return differentials (in absolute terms) than *size*. Our results visibly match this evidence.

Noteworthy, all the top predictors are relatively simple. They are derived from purely market data—such as prices, volumes, and returns. On the other hand, for example, the on-chain measures play a minor role. This signal may indicate a limited informational efficiency of the cryptocurrency markets, as investors can potentially predict returns even with simple data—easily accessible and not requiring sophisticated calculations. This conclusion is gracious for the practitioners: one can form successful predictions for the cryptocurrency markets without access to complex and expensive data, such as network activity. What matters most can be found in simple price and return data, which is much more readily available.

Fig. 3 zooms further on the ten most important variables and illustrates their relative importance. Specifically, we compare their VI across different methods. Interestingly, for many methods, most predictability is derived from a handful of key variables. This is particularly striking, for example, for LASSO, where the top three variables—*capma*, *r31_2*, and *prc*—matter more than several subsequent predictors taken together. A similar pattern is also noticeable for other linear models, as well as the tree methods, where a couple of key features (also including *illiq*, *var*, and *min*, and *r7_2*) matter incomparably more than others. The neural networks, in turn, seem more democratic; the VI appears more equally distributed among the predictors.

For robustness, in addition to the results above, we also experiment with Shapley values (see, e.g., Yun et al., 2021; Jaeger et al., 2021; Deng et al., 2023). This measure quantifies the marginal contribution of each feature (or variable) to the prediction of returns, considering all possible combinations of features. They provide a fair and consistent way to allocate the explanatory power of features. Unlike the R^2 reduction measure, Shapley values can indicate the direction of influence and account for interactions or cooperation between features. Fig. A1 in the Appendix shows the results for the OLS model. Despite the differences in methodology, the experiment yields largely consistent conclusions, with variables such as *ivol*, *skew*, *alpha*, and *r31_2* playing the essential role in prediction.

Fig. 4 sheds light on the model sparsity from a different angle. Specifically, we report the total VI for the top variables aggregated using the method of Bali et al. (2022). To this end, we first rescale all VI for each model so that their sum equals one. Then, we aggregate the scores for the top three, five, and ten predictors.

The analyses seen in Fig. 4 corroborate our earlier conclusions. The regularized regression and tree extract their information from a small number of top variables. For LASSO, the aggregate importance of the three top covariates equals about 60%, and the top ten account for >80% of the predictability. In turn, contrary to the linear regressions, the neural networks are more democratic. The aggregate VI of the top three variables does not pass even 15%, and the top ten barely crosses 60%. This implies that neural networks extract information from a broader and more balanced panel of variables than other methods. Overall, the results in Fig. 4 emphasize the remarkable dissimilarities behind the mechanics of machine learning models.

To sum up, our findings lend support to the conclusions of Liu et al. (2022). The cross-section of cryptocurrency returns is less multidimensional the commonly thought. The alleged factor zoo could be potentially reduced to a handful of simple variables, with those representing past performance and price levels being of critical importance. While other characteristics, capturing volatility or liquidity, still matter, they play a secondary role.

3.2.2. Category importance

Last, we are also interested in the contribution of different types of characteristics to overall predictability. Hence, we closely follow the approach of Bali et al. (2022) to estimate aggregate VI (rescaled previously to sum to 1) within the categories representing similar economic

intuitions—as specified in Table 1. We aim to determine which of these groups matters the most for the cross-section of cryptocurrency returns. Fig. 5 displays the results of this exercise.

The two categories that drive the prices the most are associated with past returns and liquidity. For example, in the COMB model (which averages the forecasts of other models), they account for 32 and 25% of the aggregate VI—respectively. Notably, the liquidity category also includes the *size* variable, advocated by Liu et al. (2022). The third category in the ranking is Other, driven primarily by the *prc* feature included in this class. The other measures appear less critical, with a particularly modest role played by the on-chain metrics. However, the category importance is comparably uneven and may differ substantially across the models. For example, the liquidity characteristics, which account for only 7% of the aggregate VI for ENET models, are simultaneously the most crucial set for PLS—representing 31% of the total VI. Likewise, the volatility category contribution ranges from 11% (PLS) to 21% (LASSO, ENET). This observation once again emphasizes the structural differences between various machine learning algorithms.

4. Machine learning portfolios

Having established the basic predictive properties of the machine learning signals, we now turn to portfolio analysis. We investigate whether machine learning forecasts can be transformed into effective investment strategies. We consider various study periods and investment horizons.

4.1. Portfolio construction and evaluation

Following the standard approach in asset pricing literature, we form one-way sorted portfolios. Specifically, each week, we rank all assets on their expected returns as predicted by different machine learning models. Next, we sort them into quintiles to form equal- and value-weighted portfolios.⁶ In addition, we create long-short strategies that buy (sell) the quintile of cryptocurrencies with the highest (lowest) predicted return.⁷ The performance of this portfolio provides an intuitive snapshot of the cross-sectional patterns of cryptocurrency returns.

We evaluate the portfolio performance with two distinct asset pricing models with factors derived from the cryptocurrency space. First, we employ a simple one-factor market model, which can be described as the cryptocurrency capital asset pricing model (C-CAPM):

$$r_t = \alpha_1 + \beta_{MKT} MKT_t + \varepsilon_t, \quad (6)$$

where r_t is the excess return on an examined cryptocurrency portfolio in week t ; MKT_t is the market risk factor returns, ε_t is the residual term, β_{MKT} measures the market factor exposure, and α_1 is the weekly abnormal return (the so-called Jensen's alpha). The MKT return is calculated as a value-weighted excess return on all cryptocurrencies in the sample.

The second model, advocated by Liu et al. (2022), extends model (6) with two additional factors that represent size and momentum effects:

⁶ The size distribution of the cryptocurrencies is remarkably uneven, with a handful of biggest assets accounting for most of the total market capitalization. Hence, to form more balanced yet tradeable portfolios, we cap the weight of the largest cryptocurrencies in the value-weighted portfolios. Specifically, we adopt the procedure from Jensen et al. (2022) and winsorize the market capitalization at the 90th percentile each week.

⁷ As noted by Bianchi and Babiak (2022), though short selling cryptocurrencies is not always straightforward, it can be achieved on several major exchanges—such as Binance, Bitfinex, or Poloniex. The implementation requires borrowing assets at the current market price, selling them in the market, and repurchasing later to cover the position. Alternatively, the short positions in the hedge strategies could be interpreted as relative underweighting of a relevant market benchmark.

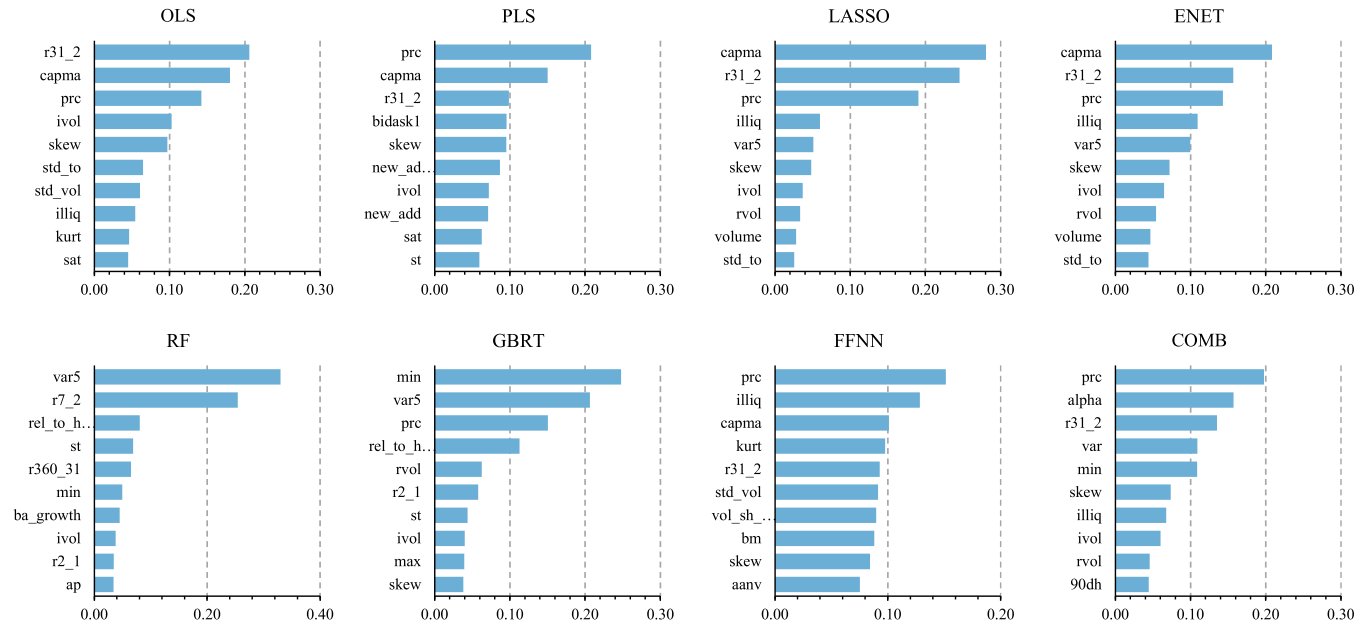


Fig. 3. Variable Importance for Different Models.

The figure presents the importance of the top 10 variables in the eight machine learning models considered in this study: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage and selection operator (LASSO), elastic net (ENET), gradient boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). The variable importance (VI) is computed as the reduction of the overall OOS R^2 resulting from excluding a given variable from the model. The panels display the reduction in R^2 from setting all values of a given variable to zero in the training sample. VI is averaged across all the training samples and rescaled to sum to 1. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018.

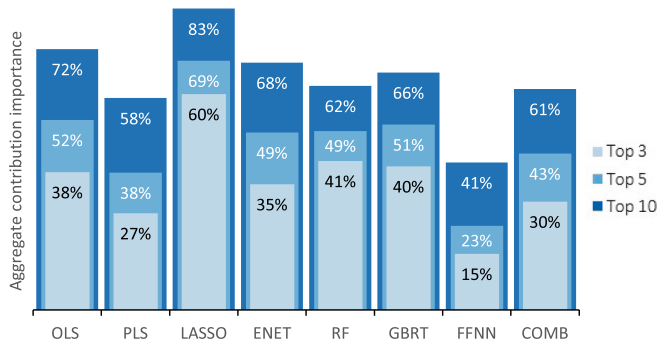


Fig. 4. Aggregate Importance of Top Predictors.

The figure illustrates the aggregate contribution of the top three, five, and ten predictors for the machine learning models used in this study: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage, and selection operator (LASSO), elastic net (ENET), gradient boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). The variable importance (VI) is computed as the reduction of the overall OOS R^2 resulting from excluding a given variable from the model. VI is averaged across all the training samples and rescaled to 1. The aggregate importance is obtained as in Bali et al. (2022). The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018.

$$r_t = \alpha_3 + \beta_{MKT}MKT_t + \beta_{SIZE}SIZE_t + \beta_{MOM}MOM_t + \varepsilon_t. \quad (7)$$

The calculation of factor returns for model (7) reproduces the procedures in Liu et al. (2022, p. 1150). Specifically, the $SIZE_t$ factor is represented by a long-short portfolio that buys (sells) 30% of cryptocurrencies with the lowest (highest) market capitalization. On the other hand, the MOM_t factor comes from 2×3 sorts on market value (size) and momentum ($r22_1$). First, we split the market into big and small cryptos by their median. Second, we determine the 30th and 70th $r22_1$ percentiles. Next, we intersect the two sorts to obtain six double-sorted

portfolios. The momentum factor is then computed as the average return on two high-momentum portfolios minus the average return on the two low-momentum portfolios.

Table A1 in the Appendix summarizes the statistical properties of returns on factor portfolios from different asset pricing models. In line with earlier studies (Liu et al., 2022), all factors generate positive average returns in our sample. Notably, all factors exhibit substantial return volatility—with the weekly standard deviation from 5.79% to 12.03%.

4.2. Baseline performance

Table 4 reports the average weekly returns on different machine learning strategies, with Panels A and B focusing on equal- and value-weighted portfolios—respectively. A quick look at the results uncovers a crucial pattern: all machine learning forecasts can be transformed into measurable economic gains for investors. All prediction models can effectively sort cryptocurrencies in line with their realized returns, so quintile portfolios systematically exhibit a monotonic or near-monotonic pattern. In consequence, all long-short strategies produce robust profits—even those that display seemingly low R^2_{OOS} values (such as FFNN).

Consider first the equal-weighted portfolios (Panel A). The mean weekly returns span from 0.88% (FFNN) to 2.35% (OLS), all statistically significant. The best-performing strategy is based on the ordinary least squares model, highlighting the differences between the cryptocurrency and equity markets. On the other hand, the tree models and neural networks fare visibly worse. In line with our earlier results from Section 3.1, accounting for non-linearities and interactions proves much less beneficial for the highly volatile and unstable cryptocurrency universe than for equities.

The machine learning profits strategies cannot be subsumed by popular risk factors. The long-short portfolios continue to generate significant alphas after accounting for factor models. Even the seminal three-factor model of Liu et al. (2022) fails to explain the machine

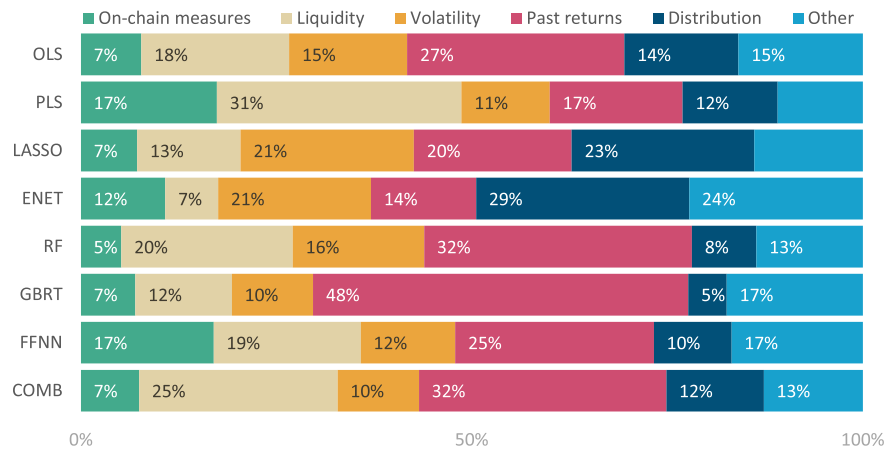


Fig. 5. Variable Importance per Category.

The figure illustrates the aggregate importance of six categories of cryptocurrency characteristics, as classified in Table 1, in terms of their overall model contribution. We consider eight machine learning models: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage and selection operator (LASSO), elastic net (ENET), gradient-boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). The variable importance (VI) is computed as the reduction of the overall OOS R^2 resulting from excluding a given variable from the model. VI is averaged across all the training samples and is rescaled to sum to 1. The VIs per category are aggregated following the method in Bali et al. (2022). The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018.

Table 4
Univariate Portfolio Sorts on Machine Learning Model Predictions.

	OLS	PLS	LASSO	ENET	RF	GBRT	FFNN	COMB
<i>Panel A: Equal-weighted portfolios</i>								
Low	0.37	0.59	0.58	0.60	0.80	0.51	0.75	0.44
2	0.41	0.79	0.62	0.59	0.92	0.61	1.04	0.79
3	0.97	0.80	1.16	1.20	0.73	1.17	1.25	0.72
4	1.15	1.56	0.74	0.70	0.86	0.97	1.03	1.35
High	2.71	1.88	2.51	2.52	2.32	2.39	1.63	2.31
High-Low	2.35	1.29	1.94	1.92	1.52	1.88	0.88	1.87
	(5.57)	(3.82)	(5.31)	(5.26)	(5.00)	(5.14)	(3.07)	(5.72)
α_1	2.32	1.31	1.95	1.93	1.58	1.91	0.87	1.89
	(6.27)	(3.78)	(5.31)	(5.28)	(4.42)	(5.12)	(2.77)	(5.39)
α_3	1.98	1.08	1.62	1.61	1.15	1.64	0.76	1.64
	(6.27)	(3.78)	(5.31)	(5.28)	(4.42)	(5.12)	(2.77)	(5.39)
<i>Panel B: Value-weighted portfolios</i>								
Low	0.35	0.50	0.45	0.46	0.78	0.61	0.18	0.28
2	0.65	0.61	0.68	0.65	0.62	0.62	0.87	0.75
3	0.69	0.93	1.19	1.22	0.63	0.79	1.43	0.76
4	1.11	1.29	0.30	0.29	0.63	0.71	0.54	0.82
High	2.51	1.51	2.83	2.87	2.46	2.86	1.74	2.65
High-Low	2.16	1.01	2.39	2.42	1.68	2.25	1.56	2.37
	(3.04)	(1.91)	(3.16)	(3.12)	(2.39)	(2.61)	(3.41)	(3.93)
α_1	2.07	0.99	2.37	2.40	1.69	2.27	1.59	2.41
	(3.33)	(1.85)	(3.22)	(3.18)	(2.35)	(2.56)	(3.02)	(3.64)
α_3	1.83	0.80	2.22	2.25	1.59	2.17	1.74	2.42
	(2.97)	(1.51)	(3.00)	(2.98)	(2.19)	(2.42)	(3.34)	(3.62)

The table presents the average weekly returns on machine learning strategies for cryptocurrency markets. We sort the cryptocurrencies into quintiles based on predictions from eight models: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage and selection operator (LASSO), elastic net (ENET), gradient boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). High (Low) is the cryptocurrency quintile with the highest (lowest) return forecast, and High-Low is the long-short strategy that buys (sells) the top (bottom) quintile. α_1 and α_3 denote alphas from the one-, and three-factor asset pricing models—respectively. All returns and alphas are reported in percentage terms. The numbers in parentheses are Newey and West (1987) t -statistics. Panels A and B report the results for equal- and value-weighted portfolios, respectively. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018.

learning returns. In consequence, α_3 on the long-short portfolios range from 1.08% (PLS) to 1.98% (OLS). To summarize, the machine learning models extract new information from cryptocurrency markets that cannot be easily captured with standard linear models.

Panel B of Table 4 reports the returns on value-weighted portfolios. The results uncover essentially similar patterns. All the long-short portfolios generate positive and highly significant profits, which any of our factor models cannot capture. The raw and abnormal returns magnitude is qualitatively similar to the equal-weighted strategies. For

example, the three-factor alphas of Liu et al. (2022) on the value-weighted portfolios range from 1.01% (PLS) to 2.42% (ENET). In a nutshell, the machine learning effectiveness works independently of the portfolio weighting scheme. This observation may matter particularly from a practical angle, as it limits the reliance on overweighting small and illiquid coins.

Notably, Table 4, Panel B, uncovers a strong performance of the COMB model—particularly on a risk-adjusted basis. The three-factor model alpha equals 2.42%, exceeding any other model's. This

observation accentuates the benefits of averaging predictions of several models. The superior profits on the COMB portfolios corroborate similar findings of Cakici et al. (2023) for international stock markets; all strategies have their benefits, but combining them proves particularly efficient.

While our discussion has so far revolved around the hedge portfolios, a closer look at the individual quintiles uncovers further insights. The cross-sectional return pattern suggests that the alphas on long-short strategies mainly originate from their long legs rather than short ones. The mean returns on the cryptocurrency quintile with the lowest expected returns are relatively moderate. For instance, for the value-weighted portfolios (Table 4, Panel B), they range from 0.19% (FFNN) to 0.5% (PLS). On the other hand, the negative returns on the top portfolios are exceptionally high: from 1.51% (PLS) to 2.86% (GBRT).

To illustrate this further, Fig. 6 displays alphas from the Liu et al. (2022) model. The cross-sectional pattern of cryptocurrency returns is more L-shaped than linear for virtually all models. The high abnormal returns of the top quintile stand out, visibly exceeding all other portfolios in absolute terms.

Notably, the substantially higher alphas of the top quintile stand in vivid contrast to the evidence from equity markets (Avramov et al., 2013; Stambaugh et al., 2012, 2015). The behavioral view on asset pricing anomalies emphasizes that buying is easier than shorting for most investors. As a result, arbitrageurs may be quicker to eliminate underpricing than overpricing. In the extreme, if short selling is severely impeded, overpricing may persist for an extended period of time. This arbitrage asymmetry can lead to higher abnormal returns on the short legs of anomaly portfolios than on the long legs.

However, the mechanism described above does not seem to be the case in cryptocurrency markets. In this regard, the return patterns in crypto assets appear to be strikingly different. Instead of being driven by short positions, the abnormal returns come almost exclusively from long positions in the top quintile. This implies that the barriers to short selling do not play a major role in the occurrence of return predictability. Thus, if the abnormal returns are due to mispricing, then the limits to arbitrage in the cryptocurrency world must be of a different nature—for example originating from liquidity constraints or trading costs. We explore this issue in more detail in Section 5.

4.3. Subperiod analysis

The return predictability in financial markets declines over time. A voluminous literature has documented that the profitability of stock characteristics has been declining due to investor learning, improvements in liquidity, or an increase in arbitrage activity (Calluzzo et al., 2019; Chordia et al., 2014; McLean & Pontiff, 2016; Schwert, 2003). Furthermore, this phenomenon extends beyond mere stocks: to currencies, industry and country portfolios, or even mutual funds (Bartram et al., 2023; Jones & Mo, 2021; Zaremba et al., 2020).⁸ This decrease in predictability typically connects with the mispricing story of asset pricing anomalies: once the market becomes more efficient, mispricing becomes less common. On the other hand, Jacobs and Müller (2020) do not confirm a robust predictability decline in international stock markets. Furthermore, Bessembinder et al. (2023) argues that factor structure depends also on the sheer quantity of assets listed in the stock market and the diversity of the firm characteristics.

Notably, the equity and cryptocurrency markets are not identical. The latter witnessed a spectacular recent growth in both the number of traded cryptocurrencies, their diversity, and market capitalization. Hence, do the abnormal returns share the same time-series patterns as in the stock market landscape? Do they gradually decline over time as cryptocurrency markets become more liquid and efficient? Or perhaps

⁸ It is important to note that Jacobs and Müller (2020) do not confirm a robust predictability decline in international stock markets.

the profitability improves, boosted by the growing cross-section of crypto assets?

Fig. 7 illustrates the cumulative returns on the long-short machine learning strategies that are considered in this study. To begin with, their performance is remarkably relatively over time—especially for the equal-weighted portfolios. No major downturns occurred within our study period, regardless of the situation in the cryptocurrency markets. The long-run fluctuations in returns are limited; however, the value-weighted portfolios reveal substantial return jumps. This indicates that the performance may be likely driven by extreme payoffs on certain currencies.

Most notably, we observe no vivid pattern of declining portfolio returns over time. Indeed, some strategies perhaps witness a minor flattening of the profit curve over the last year, but the tendency is far from evident. In general, all strategies continue to generate profits throughout the entire study period.

To scrutinize this effect further, we reproduce our portfolio sorts from Table 4 in subperiods. Specifically, we split the whole research period into rough halves (March 8, 2018, to August 19, 2020, and August 20, 2020, to January 26, 2023) and check the performance therein. Table 5 reports the results of this exercise. For brevity, we limit the presentation to value-weighted strategies.⁹

Both subperiods exhibit substantial return predictability. Mean returns and alphas are large and positive for all models in both sample halves. Most models show a certain decrease in profitability in the latter sample, but the effect is rather weak. More specifically, six out of eight models display lower returns in the second subperiod than in the first. The size of these differences, however, is not consistent. While the three-factor alpha on ENET falls from 2.96% in the first subperiod to 1.39% in the second, the average abnormal return on PLS actually increases from 0.71% to 1.14%. The alpha of the comprehensive COMB model shrinks only slightly, from 2.46% to 2.34%.

In summary, our findings from cryptocurrency markets differ from previous evidence from equities and other asset classes. We do not observe a sizeable systematic decline in return predictability. The forces that reduce mispricing in the equity universe are apparently absent in the cryptocurrency world. In fact, the efficiency and liquidity of the US stock market have systematically improved over the past decades. Meanwhile, cryptocurrencies have experienced a proliferation of small and illiquid coins, which may constitute reservoirs of return predictability. Perhaps the maturation of cryptocurrency markets will reduce abnormal returns in the future, but we do not observe this yet.

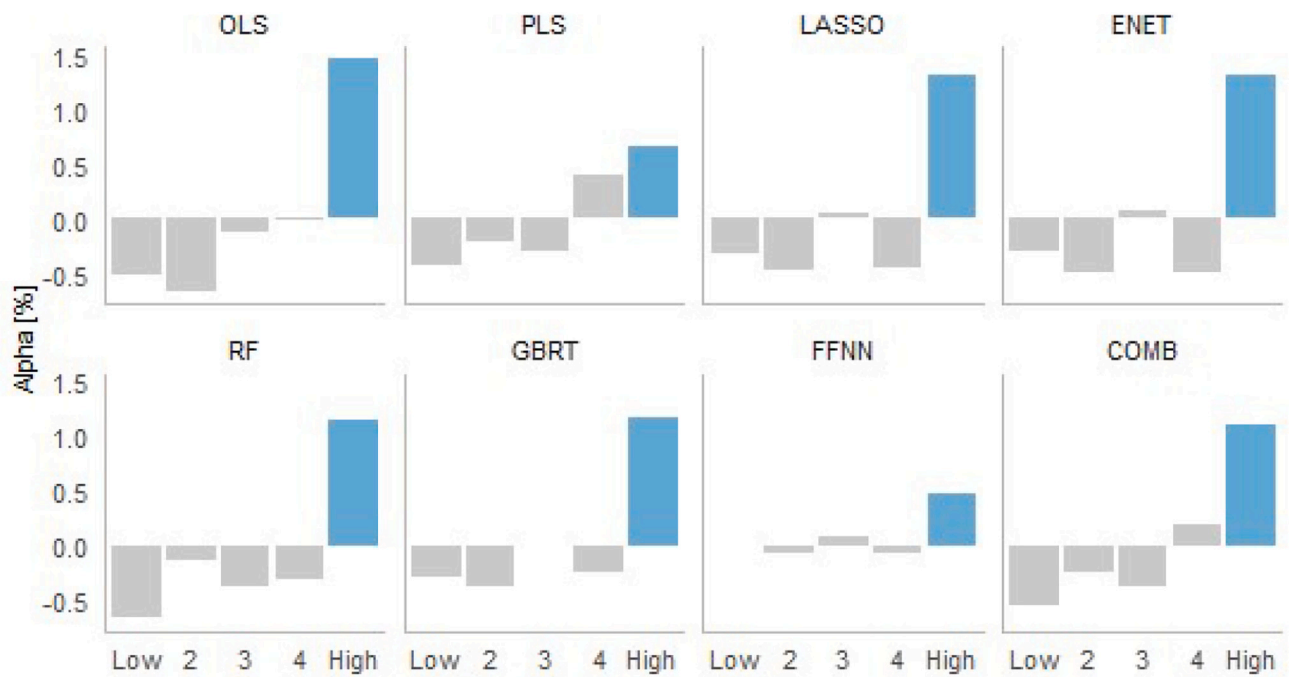
4.4. Extended forecast horizon

Our tests so far have concentrated on a one-week prediction horizon. However, return predictability with machine learning may vary substantially across different horizons. For example, Coqueret (2022) documents that predictability is typically stronger for longer forecast horizons. Consistently with this, Gu et al. (2022) and Leippold et al. (2022) observe higher R^2_{OOS} for longer horizons. The effect may be associated with the higher signal-to-noise ratio in long-term returns and more pronounced signal persistence. Does the prediction horizon also matter for machine learning forecasts in cryptocurrencies?

The relatively short data availability for the cryptocurrency markets does not allow testing excessively long forecast horizons. Using monthly or quarterly data would substantially reduce the number of distinct (non-overlapping) observations in the sample. Nonetheless, to shed light on the issues above, we reproduce our baseline analysis from Section 4.2 using a two-week forecast horizon. To verify this, we revisit our univariate portfolio sorts from Section 4.2 with a modified prediction horizon and holding period. Specifically, we train the models using the 14-

⁹ The results for equal-weighted portfolios are available upon request.

Panel A: Equal-weighted portfolios



Panel B: Value-weighted portfolios

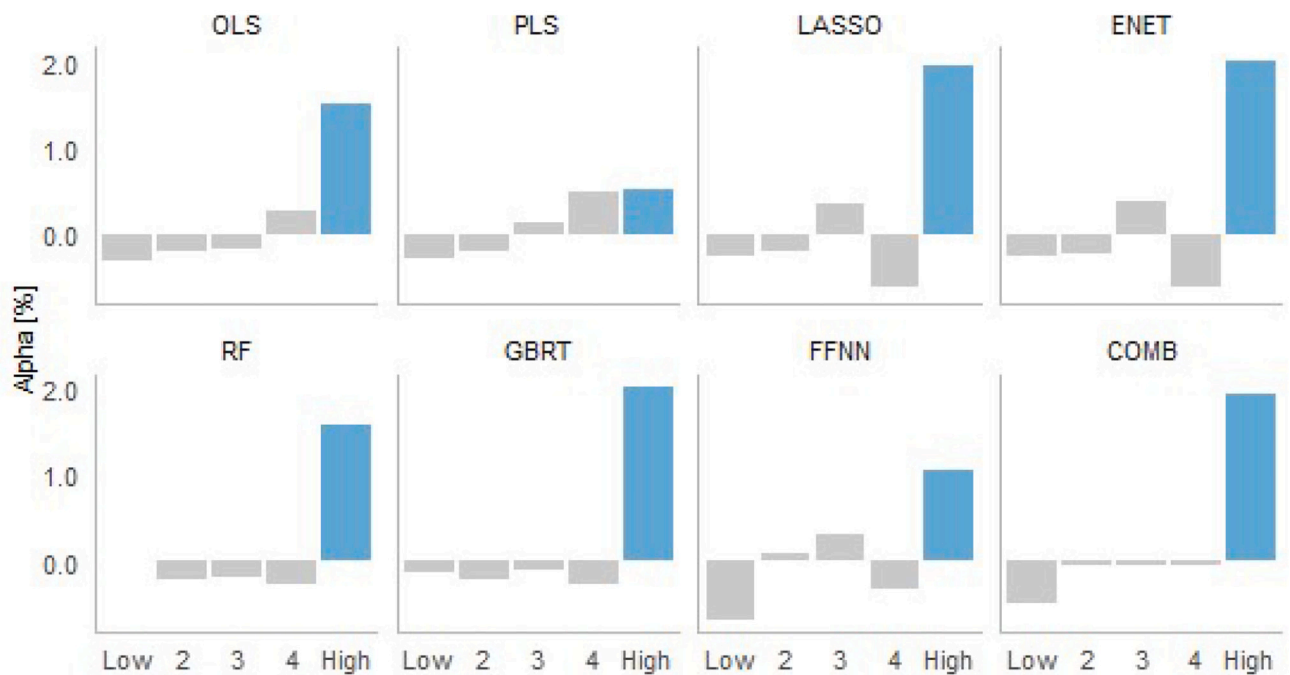
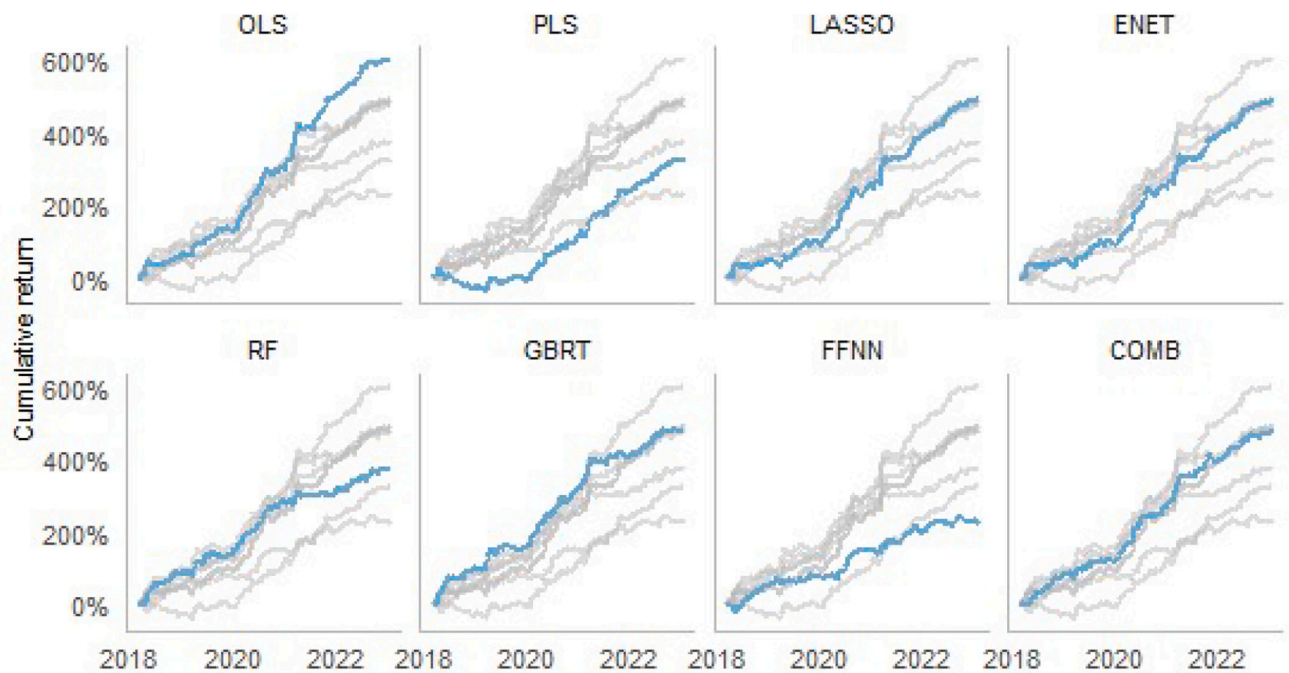


Fig. 6. Abnormal Returns on Quintile Machine Learning Portfolios.

The figure presents the weekly alphas from the three-factor model of Liu et al. (2022) on quintile portfolios formed on machine learning forecasts. We sort the cryptocurrencies into quintiles based on predictions from eight models: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage and selection operator (LASSO), elastic net (ENET), gradient boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). High (Low) is the cryptocurrency quintile with the highest (lowest) return forecast. The blue bars indicate the portfolios with the lowest expected returns. All alphas are reported in percentage terms. Panels A and B report the results for equal- and value-weighted portfolios, respectively. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Panel A: Equal-weighted portfolios



Panel B: Value-weighted portfolios

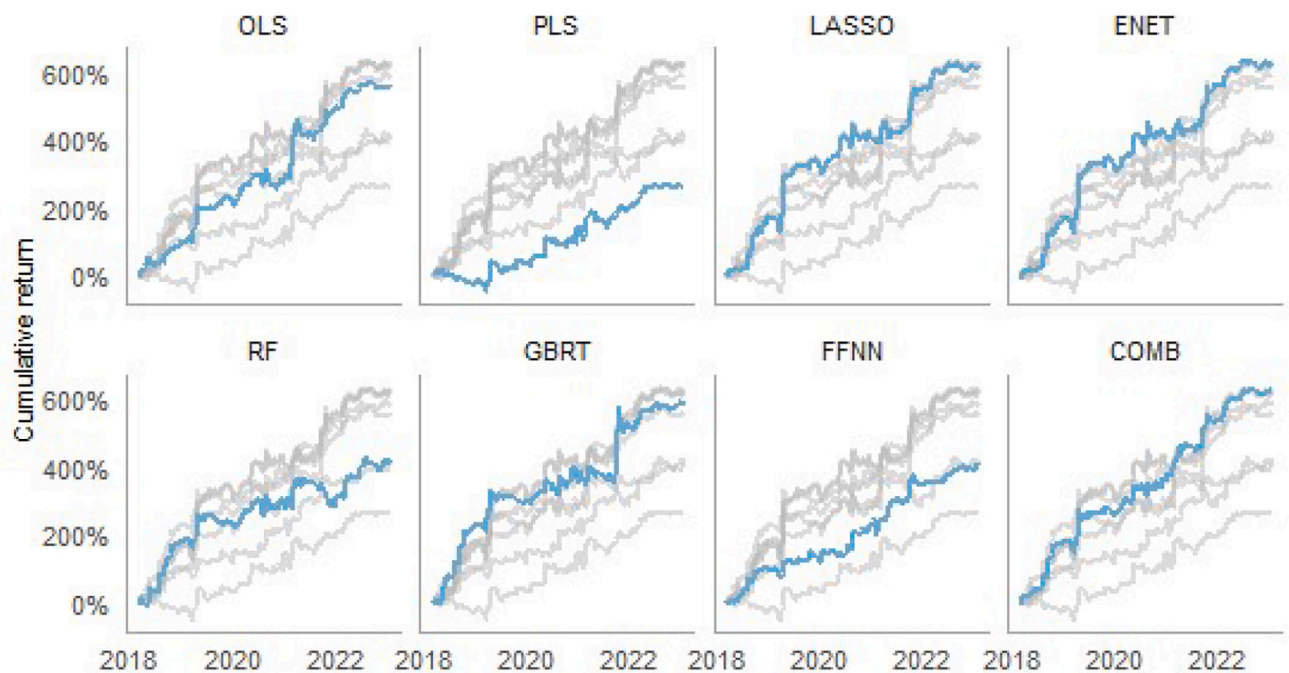


Fig. 7. Cumulative Returns on Long-Short Machine Learning Portfolios.

The figure presents the cumulative returns on long-short machine learning portfolios. We sort the cryptocurrencies into quintiles based on predictions of eight models: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage and selection operator (LASSO), elastic net (ENET), gradient boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). The reported strategies assume a long (short) position in the quintile of cryptocurrencies with the highest (lowest) expected return. The strategies are rebalanced weekly. Panels A and B report the results for equal- and value-weighted portfolios, respectively. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018. The blue line represents the strategy indicated in the headline, and the grey lines display other strategies for comparison. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5
Returns on Machine Learning Portfolios in Subperiods.

	OLS	PLS	LASSO	ENET	RF	GBRT	FFNN	COMB
<i>Panel A: First half (8 March 2018 to 19 August 2020)</i>								
Low	−0.02	−0.01	−0.09	−0.09	0.03	−0.12	−0.28	−0.22
2	−0.22	0.11	−0.06	−0.06	−0.10	0.05	0.09	−0.12
3	−0.04	0.01	0.20	0.21	0.15	0.14	0.24	0.42
4	0.53	0.71	−0.13	−0.12	−0.16	−0.03	0.25	0.25
High	2.11	0.76	2.95	2.99	2.24	2.37	0.94	2.28
High-Low	2.12	0.77	3.04	3.09	2.21	2.48	1.22	2.50
	(2.66)	(0.99)	(2.49)	(2.44)	(2.11)	(2.50)	(2.03)	(2.62)
α_3	2.07	0.71	2.91	2.96	2.17	2.36	1.38	2.46
	(2.97)	(1.51)	(3.00)	(2.98)	(2.19)	(2.42)	(3.34)	(3.62)
<i>Panel B: Second half (20 August 2020 to 26 January 2023)</i>								
Low	0.72	1.00	0.98	1.01	1.54	1.33	0.64	0.77
2	1.52	1.10	1.42	1.35	1.34	1.18	1.64	1.63
3	1.41	1.85	2.19	2.24	1.12	1.45	2.62	1.10
4	1.68	1.87	0.74	0.70	1.42	1.45	0.82	1.39
High	2.91	2.25	2.72	2.75	2.69	3.35	2.54	3.01
High-Low	2.19	1.25	1.73	1.74	1.14	2.03	1.91	2.25
	(1.84)	(1.76)	(1.96)	(1.97)	(1.20)	(1.42)	(2.64)	(2.87)
α_3	1.80	1.14	1.38	1.39	1.12	1.82	2.22	2.34
	(1.94)	(1.46)	(1.49)	(1.51)	(1.08)	(1.20)	(2.61)	(2.68)

The table presents the average weekly returns on machine learning strategies for cryptocurrency markets. We sort the cryptocurrencies into quintiles based on predictions from eight models: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage and selection operator (LASSO), elastic net (ENET), gradient boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). High (Low) is the cryptocurrency quintile with the highest (lowest) return forecast, and *High-Low* is the long-short strategy that buys (sells) the top (bottom) quintile. α_3 denotes alphas from the three-factor asset pricing models. All returns and alphas are reported in percentage terms. The numbers in parentheses are Newey and West (1987) *t*-statistics. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018. Panels A and B report the results for the first and second half of the full testing period, i.e., March 8, 2018, to August 19, 2020 and August 20, 2020, to January 26, 2023.

day returns and use their predictions to form bi-weekly reconstructed portfolios. Table 6 reports the returns on machine learning portfolios implemented using the aforementioned procedures. Again, for conciseness, we concentrate on value-weighted portfolios.

As seen in Table 6, machine learning signals are generally short-lived. Using a longer-horizon predictions results in a noticeable decline in machine learning profitability. The mean weekly long-short portfolio returns range from 0.83% (FFNN) to 1.85% (OLS), being, on average, 29% lower than in the case of weekly forecasts. A similar drop is observable for alphas. For example, the three-factor alpha on the comprehensive COMB strategy decreases from 2.42% for weekly portfolios to 1.31% for the bi-weekly ones. This pattern marks another remarkable difference when compared to equity markets. The investors are not better off from long-horizon forecasts, as Coqueret (2022) suggested. On the contrary, the trading signals seem to be rather short-term in nature. As a result, extending the forecast horizon and reducing portfolio rebalancing frequency materially impedes machine learning profitability.

Table 6
Machine Learning Strategies for an Extended Holding Period.

	OLS	PLS	LASSO	ENET	RF	GBRT	FFNN	COMB
Low	0.11	0.29	0.33	0.32	0.75	0.28	0.24	0.27
2	0.60	0.75	0.83	0.87	0.67	0.64	0.90	0.59
3	0.53	0.73	1.05	1.03	0.60	0.88	0.88	0.83
4	1.32	1.13	0.48	0.45	0.75	0.79	0.89	0.85
High	1.96	1.79	1.73	1.73	1.42	1.65	1.07	1.77
High-Low	1.85	1.50	1.40	1.41	0.67	1.37	0.83	1.50
	(2.83)	(1.93)	(1.87)	(1.89)	(0.93)	(2.31)	(1.27)	(1.97)
α_3	1.60	1.30	1.22	1.23	0.59	1.37	0.90	1.31
	(3.13)	(2.30)	(1.96)	(1.98)	(0.85)	(2.28)	(1.53)	(2.14)

The table presents the average weekly returns on machine learning strategies for cryptocurrency markets. We sort the cryptocurrencies into quintiles based on predictions from eight models: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage and selection operator (LASSO), elastic net (ENET), gradient boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). All the models are trained to predict two-week returns, and, accordingly, the portfolios are reconstructed bi-weekly. High (Low) is the cryptocurrency quintile with the highest (lowest) return forecast, and *High-Low* is the long-short strategy that buys (sells) the top (bottom) quintile. α_1 and α_3 denote alphas from the one-, and three-factor asset pricing models—respectively. All portfolios are value-weighted, and the returns and alphas are reported in percentage terms. The numbers in parentheses are Newey and West (1987) *t*-statistics. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018.

Table 7

Research Design Choices.

Design choice	Options
<i>Panel A: Sample preparation</i>	
Minimum market capitalization	1, 2, 5, 10 million USD
Minimum share price	none, \$0.10, \$0.20
Minimum trading history	none, 13 weeks, 26 weeks
Return winsorization	none, 0.5th/99.5th pct., 1st/99th pct.
<i>Panel B: Portfolio construction</i>	
Number of portfolios	3, 4, 5, 10
Minimum number of assets	1, 5, 10, 20
Weighting scheme	Equal-, value-, capped-value weighted

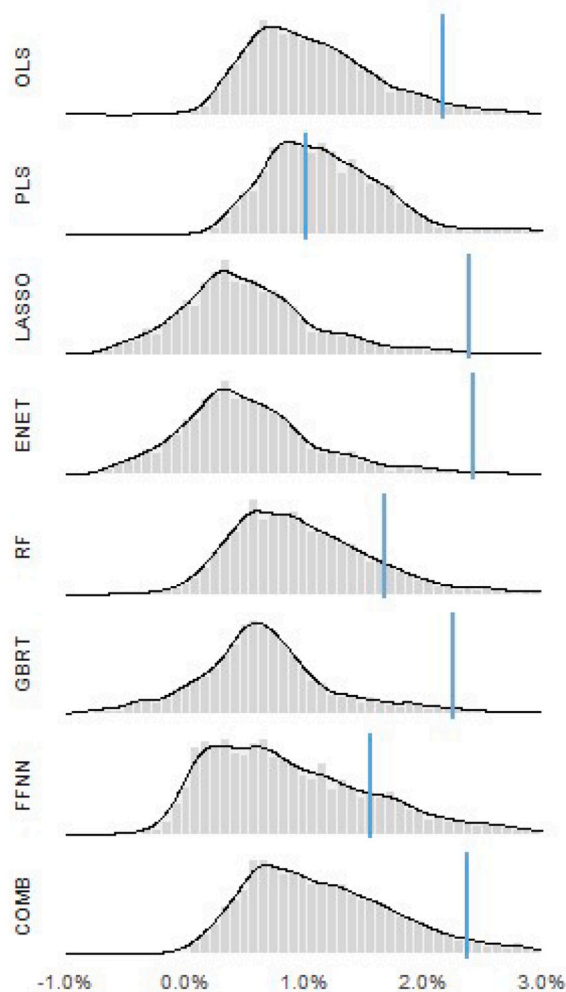
The table presents the research design concerning sample preparation and portfolio construction used to analyze non-standard errors (Section 4.5). The combinations of all possible design choices yield 5184 possible variants.

regardless of their price and trading history. In this robustness check, we relax these conditions in several ways. More precisely, we consider four different thresholds for market capitalization (\$1, \$2, \$5, or \$10 million), three filters for minimum market price (\$0.20, \$0.10, or all), and three approaches for minimum trading history (13 weeks, 26 weeks, or all). We also incorporate three additional winsorization overlays: at the 0.5th and 99.5th percentiles, the 1st and 99th percentiles, or none.

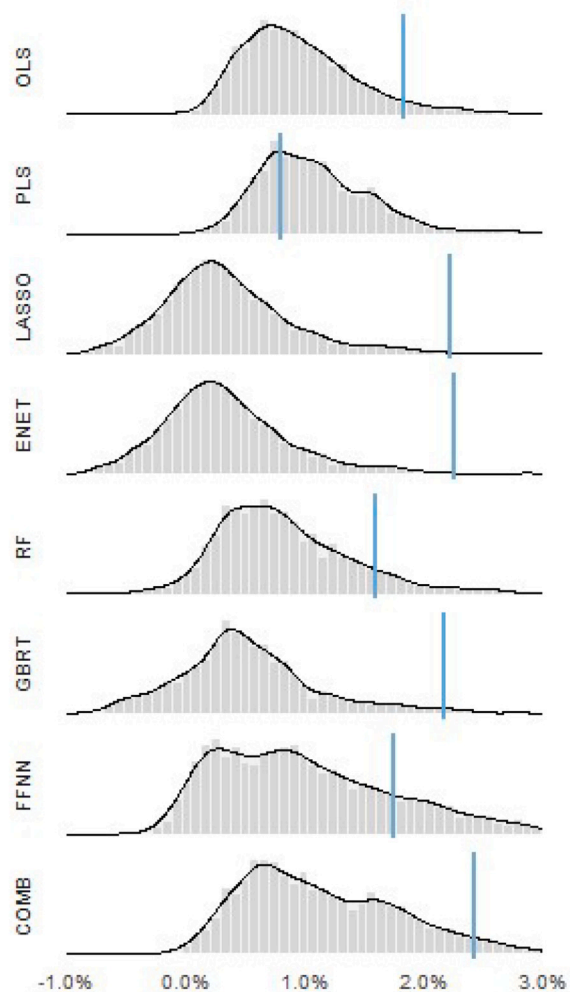
Regarding factor portfolio construction, our initial framework adopts quintile value-weighted portfolios with no minimum number of assets and uses an equal or capped value-weighting scheme. We now reassess each of these choices: we examine four different breakpoints (terciles, quartiles, quintiles, and deciles), four minimum asset numbers (1, 5, 10, or 20), and three possible weighting schemes: equal-weighting, value-weighting, and capped value-weighting (with equity winsorized at the 90th percentile).

In total, the combinations of all possible decision nodes yield 5184 alternative research designs. Some of these may be infeasible because a

Panel A: Mean returns



Panel B: Three-factor model alphas

**Fig. 8.** Alternative Research Designs.

The figure illustrates the mean returns (Panel A) and three-factor model alphas (Panel B) on various implementations of machine learning strategies for cryptocurrency markets. Using machine learning predictions as inputs, we consider seven decision nodes regarding the strategy design: minimum market capitalization (1, 2, 5, or 10 million USD), minimum coin market price (\$0.20, \$0.10, or all), minimum trading history (13 weeks, 26 weeks, or all), return winsorizing (0.5/99.5th percentile, 1/99th percentile, or none), type of portfolio sorts (terciles, quartiles, quintiles, or deciles), the minimum number of assets per portfolio (1, 5, 10, or 20), and portfolio weighting scheme (equal-weighted, value-weighted, capped value-weighted). Combining all possible methodological decisions generates 4923 portfolio designs with at least 52 weeks of history, whose performance is reported by the density plots below. The blue lines indicate our baseline results for value-weighted from Table 4. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018. All values are reported in percentage terms. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

combination of several criteria may prove too stringent. For example, requiring a high minimum security price, a large market capitalization, and a long trading history simultaneously may leave too few assets available to construct decile portfolios of 10 or 20 coins—or at least not over the entire study period. Therefore, out of the 5184 variants, we consider only those with at least 52 weeks of history available, reducing the number of combinations examined to 4923.

Notably, most alternative study designs are more conservative than our baseline approach. Most impose stricter minimum prices, market capitalization, or trading history filters. Thus, if return predictability is attributable to difficult-to-trade assets—as in the stock market (Avramov et al., 2023)—the typical profitability across alternative designs is likely to be lower.

Figure 8 illustrates the distributions of the performance measures for the various implementation of our machine learning strategies. Overall, their performance proves robust across alternative research designs. The vast majority of specifications generate positive mean returns and alphas for all models. Interestingly, some models prove more robust than others, performing noticeably better across different implementations. For example, the COMB's distribution signals relatively high returns compared to other models, such as LASSO and ENET. The robustness of the combination model may stem from the built-in averaging mechanism. While individual models may “overfit” a particular research design, e.g., by extracting profitability from a specific type of stock, the COMB approach overcomes this problem at least partly by minimizing its reliance on model-specific assumptions.

The results in Figure 8 come with a caveat, however. For most models, the majority of research designs underperform our baseline methodology employed in Section 4.2. In other words, the mean returns and alphas of our long-short machine learning strategies typically visibly beat most alternative implementations. The effect is particularly visible for LASSO and ENET, where our baseline approach is among the very best research designs out of all 4923 considered. Only PLS seems to escape this pattern, with most alternative designs performing better than those in Table 4.

Why is the performance of most alternative designs inferior to our baseline methodology? To address this question, we analyze the impact

of each decision node on portfolio returns. To be specific, we hold one specific decision constant and examine the return differences across all other research designs. For example, to examine the impact of the \$10 million capitalization filter, we look at all 1296 [=5184/4] research designs with this assumption and compare their average performance with other size filters. This methodology allows us to pinpoint the impact of particular decisions on strategy returns.

Fig. 9 showcases the results for the comprehensive COMB strategy, which aggregates forecasts of all individual models. An overview of mean returns across different methodological choices reveals several salient patterns. First, the number of portfolios in the cross-section matters. The payoffs are substantially higher for long-short strategies based on deciles than on quartiles or terciles. The decile portfolios capture a broader spread between the top and bottom return forecasts, which, intuitively, results in higher return differentials. Second, also the weighting scheme plays a substantial role: equal-weighted portfolios visibly beat their value-weighted counterparts. Weighting cryptocurrencies equally overemphasize small and volatile coins with extreme returns, while value-weighting diminishes their importance. Higher returns on equal-weighted portfolios may signal the critical role of such assets.

In addition to the above, two further factors that matter are winsorizing and limits on firm size. The portfolio performance appears highly sensitive to the threshold adopted to winsorize the extreme returns. Shrinking their values visibly decreases portfolio profitability. This implies that machine learning profits depend critically on harvesting extreme returns on volatile cryptocurrencies. Reducing the impact of outliers markedly dampens portfolio performance. Finally, the impact of the market capitalization filter is somewhat smaller yet still visible: excluding the smallest coins reduces the return differentials.

In summary, analyzing non-standard errors sheds new light on sources of abnormal returns on machine learning strategies. The forecasting models extract profitability from extreme price movements on small and volatile coins. Hence, limiting their role—either through portfolio weights, return winsorization, or filtering our smallest cryptocurrencies—reduces portfolio profitability. Consequently, our baseline approach typically underperforms most of the alternative—and

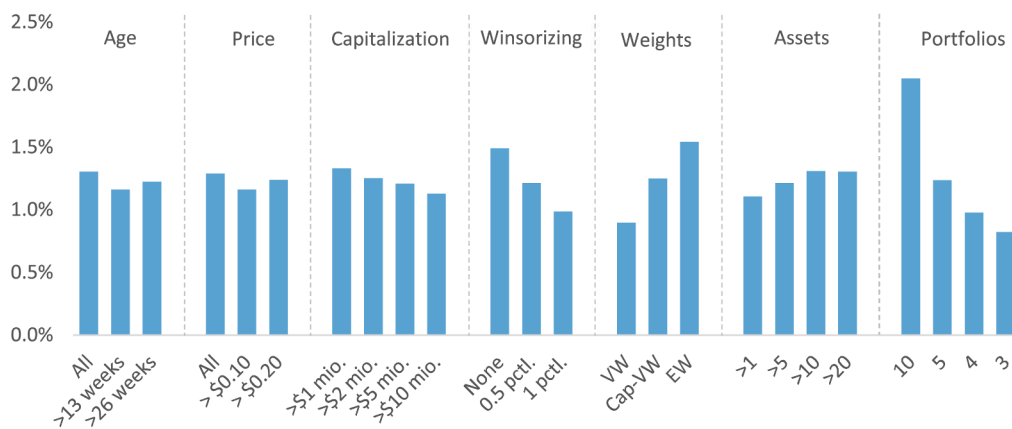


Fig. 9. Machine Learning Performance across Different Methodological Choices.

The figure illustrates the averages of mean weekly returns on the long-short machine learning portfolios formed using the combination forecast (COMB) model for across various research designs. The strategies buy the cryptocurrencies with the highest (lowest) predicted returns and are rebalanced weekly. Using COMB predictions as inputs, we consider seven decision nodes regarding the strategy design: minimum market capitalization (1, 2, 5, or 10 million USD), minimum coin market price (\$0.20, \$0.10, or all), minimum trading history (13 weeks, 26 weeks, or all), return winsorizing (0.5/99.5th percentile, 1/99th percentile, or none), type of portfolio sorts (terciles, quartiles, quintiles, or deciles), the minimum number of assets per portfolio (1, 5, 10, or 20), and portfolio weighting scheme (equal-weighted, value-weighted, capped value-weighted). Combining all possible methodological decisions generates 4923 portfolio designs with at least 52 weeks of history. The portfolios are fanned out over the decision nodes indicated on the horizontal axis so that we present averages of the mean returns of all of the 4923 portfolios that are formed using the specific methodological choice. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018. All values are reported in percentage terms.

typically more conservative—research design choices.

Our discussion thus far has concentrated on the predictions of the COMB model. Table 8, in turn, extends the analysis to all individual models as well. The general tendencies from Fig. 9 prevailingly apply to most individual models. However, a closer look also reveals some interesting differences. For example, specific models such as OLS, LASSO, and ENET depend considerably on the minimum market price level. Discarding the low-priced coins measurably reduces portfolio profitability. Likewise, the impact of portfolio weights is not identical across all strategies. Finally, Table 8 also confirms the remarkable robustness of the COMB model. Being particularly resilient to methodological changes, this strategy commonly outperforms most—if not all—peers.

5. Return predictability and limits to arbitrage

Machine learning methods aggregate multiple signals from individual anomalies. Accordingly, any weaknesses of these signals may transmit into the models' forecasts. This inference may have substantial consequences for machine learning effectiveness. For example, in the equity universe, most well-known anomalies—such as momentum, value, or profitability—extract their alphas mainly from small and illiquid firms (Cakici & Zaremba, 2022; Hong et al., 2000; Hou et al., 2020; Israel & Moskowitz, 2013). This evidence aligns with the view that mispricing typically arises where arbitrage activities are constrained. In line with this, as shown by Avramov et al. (2022), machine learning profits also tend to concentrate on difficult-to-arbitrage stocks.

Return predictability is commonly attributed to one of two competing explanations: risk or mispricing. If the latter is true, it should prevail in difficult-to-arbitrage assets. So, does it? We run two experiments to shed light on these issues: cross-sectional regressions with

interactions and bivariate portfolio sorts.

5.1. Cross-sectional regressions with interaction terms

To identify the difficult-to-trade securities, we employ three popular proxies of limits of arbitrage: idiosyncratic volatility (*ivol*), illiquidity (*illiq*), and bid-ask spread (*bidask*). Numerous asset pricing studies have argued that assets with values of these variables are costly to trade and difficult to hedge; consequently, any mispricing is likely to persist (e.g., Pontiff, 2006; Sadka & Scherbina, 2007; Chordia et al., 2008; McLean, 2010; Lam & Wei, 2011). In addition to these three variables, we also calculate an aggregate measure of limits of arbitrage (*lim*). For each cryptocurrency, it is represented by its average cross-sectional rank on *ivol*, *illiq*, and *bidask* in a given week.

To explore the impact of the limits to arbitrage on return predictability, we estimate weekly cross-sectional regressions with interaction terms accounting for the limits to arbitrage:

$$r_{i,t+1} = \gamma_{0,t} + \gamma_{1,t}E_t(r_{i,t+1}) + \gamma_{2,t}Arb_{i,t} + \gamma_{3,t}E_t(r_{i,t+1})Arb_{i,t} + \gamma_{k,t}k_{i,t} + \varepsilon_{i,t+1}, \quad (8)$$

where $E_t(r_{i,t+1})$ is the cryptocurrency i expected return, as predicted by the COMB model, and $r_{i,t+1}$ is its realized value. For brevity, we limit our presentation to the COMB predictions, which aggregates all individual models; using other models yields qualitative results and is available upon request. $Arb_{i,t}$ is a dummy variable taking a value of one if a given measure of limits to arbitrage (*ivol*, *illiq*, *bidask*, *lim*) is higher than the weekly cross-sectional median or zero otherwise. k is the vector of additional control variables, accounting for market beta, size, and momentum effects. $\varepsilon_{i,t+1}$ is the error term and $\gamma_{0,t}$, $\gamma_{1,t}$, $\gamma_{2,t}$, $\gamma_{3,t}$, and $\gamma_{k,t}$ are the estimated weekly regression coefficients. Our focus is on the

Table 8
Average Portfolio Returns Across Different Research Designs.

		OLS	PLS	LASSO	ENET	RF	GBRT	FFNN	COMB
Age	All	1.07	0.99	0.57	0.56	0.93	0.75	1.11	1.30
	>13 weeks	1.05	1.27	0.51	0.50	0.97	0.63	0.86	1.16
	>26 weeks	1.20	1.41	0.54	0.53	1.10	0.72	0.87	1.22
Price	All	1.49	1.21	0.69	0.69	1.01	0.93	0.79	1.29
	> \$0.10	0.97	1.23	0.43	0.43	1.07	0.68	0.98	1.16
	> \$0.20	0.85	1.23	0.48	0.47	0.91	0.49	1.08	1.24
Capitalization	>\$1 million	1.21	1.21	0.64	0.63	1.10	0.79	0.96	1.33
	>\$2 million	1.14	1.21	0.58	0.57	1.01	0.72	0.93	1.25
	>\$5 million	1.14	1.24	0.50	0.50	1.03	0.70	0.96	1.21
	>\$10 million	0.94	1.24	0.43	0.42	0.86	0.61	0.93	1.13
Winsorizing	None	1.29	1.29	0.76	0.76	1.26	0.95	1.07	1.49
	0.5th percentile	1.11	1.22	0.53	0.52	0.98	0.69	0.95	1.21
	1st percentile	0.92	1.16	0.32	0.32	0.76	0.47	0.82	0.99
Weights	Value-weighted	1.24	1.07	0.61	0.60	1.01	0.86	0.34	0.90
	Cap value-weighted	1.18	1.18	0.69	0.68	1.09	0.71	1.06	1.25
	Equal-weighted	0.91	1.41	0.31	0.31	0.89	0.54	1.44	1.54
Assets	>1	1.15	1.35	0.55	0.54	1.08	0.78	0.75	1.11
	>5	1.05	1.22	0.54	0.53	0.98	0.74	0.95	1.21
	>10	1.06	1.22	0.55	0.54	1.00	0.66	1.04	1.31
	>20	1.17	1.08	0.51	0.51	0.93	0.61	1.08	1.31
Portfolios	10	1.71	1.89	1.21	1.19	1.47	1.36	1.62	2.05
	5	1.10	1.20	0.44	0.43	1.06	0.68	1.01	1.24
	4	0.94	1.08	0.30	0.30	0.89	0.56	0.70	0.98
	3	0.80	0.87	0.33	0.34	0.67	0.35	0.59	0.82

The table illustrates the averages of mean weekly returns on the long-short machine learning portfolios for across various research designs. The strategies buy the cryptocurrencies with the highest (lowest) predicted returns and are rebalanced weekly. We use eight prediction models: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage and selection operator (LASSO), elastic net (ENET), support vector machine (SVM), gradient boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). Using the models' forecasts as inputs, we consider seven decision nodes regarding the strategy design: minimum market capitalization (1, 2, 5, or 10 million USD), minimum coin market price (\$0.20, \$0.10, or all), minimum trading history (13 weeks, 26 weeks, or all), return winsorizing (0.5/99.5th percentile, 1/99th percentile, or none), type of portfolio sorts (terciles, quartiles, quintiles, or deciles), the minimum number of assets per portfolio (1, 5, 10, or 20), and portfolio weighting scheme (equal-weighted, value-weighted, capped value-weighted). Combining all possible methodological decisions generates 4923 portfolio designs with at least 52 weeks of history. The portfolios are fanned out over the decision nodes indicated on the horizontal axis so that we present averages of the mean returns of all of the 5184 portfolios that are formed using the specific methodological choice. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018. All values are reported in percentage terms.

coefficient $\gamma_{3,t}$; its positive value would imply that high limits to arbitrage augment the return predictability by machine learning models.

Table 9 reports the average slope coefficients from regression (8). The limits to arbitrage clearly matter. The $\gamma_{3,t}$ coefficients are positive and highly significant in all specifications, regardless of the particular proxy used. Admittedly, the coefficients on $E_t(r_{i,t+1})$ also remain significant in all cases (at least at the 10% level); this signifies that machine learning models continue to forecast returns successfully even if arbitrage constraints are controlled for. In other words, they work even in market segments where trading is relatively more straightforward. Nonetheless, the return predictability in difficult-to-trade assets is measurably stronger. This observation complies with the hypothesis that return predictability in the cryptocurrency markets originates mainly from mispricing.

Notably, specifications (1) to (5) in Table 9 include simple regressions that only consider the limits to arbitrage and the associated interaction effects, while specifications (6) to (9)—for robustness—include additional control variables. These include the well-established cryptocurrency returns underlying Liu et al.'s (2022) three-factor model: market beta, size, and momentum. Importantly, our conclusions hold consistently across regressions. In fact, the cross-sectional predictability of the control variables is limited because their information content is already extracted in the machine learning forecasts.

5.2. Bivariate portfolio sorts

We carry on with bivariate portfolio sorts to corroborate the conclusions from the cross-sectional regressions seen in Section 5.1. In this analysis—each week—we rank cryptocurrencies on one of the four proxies for arbitrage constraints, *ivol*, *illiq*, *bidask*, and *lim*, and sort them into terciles. Thus, we generate three market segments with high, medium, and low arbitrage difficulties. Next, within each of these segments, we sort the assets again on their predicted returns from a machine learning model. Again, for conciseness, we limit the presentation to the COMB model.¹⁰ Next, we weight the cryptocurrencies—equally or on their capitalization—in order to obtain nine portfolios from double-dependent sorts. Finally, we calculate long-short machine learning strategies within each of the terciles representing different levels of limits to arbitrage. Table 10 presents the outcomes of this analysis.

Table 10, Panel A reports the average portfolio returns. Overall, the machine learning strategies work effectively across all market segments. The long-short portfolios generate positive and significant mean raw and abnormal returns in each of the considered specifications. Nonetheless, their magnitude evidently differs. The payoffs are considerably more solid in the market segments with high barriers to arbitrage than in those when trading is more straightforward.

Take the aggregate *lim* proxy for arbitrage constraints as an example (Table 10, Panel A.1). The equal-weighted long-short strategy (left section of Panel A) earns an average return of 2.54% (t -stat = 5.08) per week in the high *lim* tercile. On the other hand, in the low arbitrage segment, the equivalent mean return amounts to 0.73% (t -stat = 2.06). The difference between the high and low *lim* segments equals 1.81% and is highly significant (t -stat = 2.89). Furthermore, it cannot be explained by common risk factors from the model of Liu et al. (2022). Moreover, the value-weighted portfolios yield similar results, as seen in the left section of Panel A.

Panels A.2 to A.4 of Table 10 shed further light on the role of different components of the *lim* measure. Which proxies for the limits to arbitrage matter the most? The strongest performance dispersion is

observed for *ivol*. Consider the equal-weighted portfolios again. In this case, the average difference between long-short portfolio alpha in the high and low *ivol* terciles equals 2.64% (t -stat = 4.52). The difference-in-difference return is even bigger for the value-weighted strategies, amounting to 3.15% (t -stat = 2.37). On the other hand, *illiq*'s effect is relatively weaker—though still visible. The alpha difference between the equal-weighted (value-weighted) high and low *bidask* tercile amounts to 0.88% (1.08%).

Our earlier evidence in Section 4.2 indicates that most of the abnormal returns originate from the long legs of investment strategies. A closer look at individual bivariate portfolios in Table 10, Panel A leads to similar conclusions. However, the magnitude of returns is not equal across different market segments. Concretely, it is powerful in terciles with high limits to arbitrage and relatively weaker in terciles with low constraints. Consider value-weighted portfolios from two-way sorts on *lim*. The average returns on the cryptocurrencies with the lowest expected returns span between −0.05% and 0.41%; on the other hand, the ones with the highest predictions are between 1.44 and 3.53%. Notably, the most outstanding mean return of 3.53% is recorded precisely on the portfolio with the highest *lim* values. In other words, the machine learning strategies derive their predictability—particularly from long positions in difficult-to-arbitrage securities.

The conclusions above are especially interesting when juxtaposed with the average cryptocurrency capitalization in different market segments and their respective relative size. Table 10, Panel B uncovers the details of such distribution. Consider the bivariate sorts on expected returns in *lim* in Panel B.1. The tercile of cryptocurrencies with the lowest limits to arbitrage (the one when machine learning profits are the weakest) accounts for 82.88% of the cryptocurrency market capitalization. Meanwhile, the tercile of the most difficult-to-trade coins (the ones which generate the highest profits) represents only 1.82% of the market value. In other words, the market segment where the return predictability is the most robust accounts for a minuscule fraction of the cryptocurrency universe.

Furthermore, the two-way sorted portfolio with the highest expected return in the high *lim* tercile—which produced a remarkable mean value-weighted return in Panel B of −3.58%—represents only 0.20% of the market value, and the average cryptocurrency capitalization therein equals \$60 million. Consequently, trading such small cryptocurrencies effectively may be challenging. Moreover, their economic significance is relatively modest.

To conclude, the machine learning models work across diverse segments of cryptocurrency markets. Nevertheless, the return predictability is the strongest where trading is the most difficult. Particularly, it originates from long positions in difficult-to-trade assets—which aggregate market value; because of this, economic importance may be limited.

6. Practical investor perspective

Finally, we are also interested in the practical perspective on return predictability with machine learning. Do these forecasts translate into feasible strategies? Can they survive real-world conditions? To throw light on these issues, we look closer at three aspects: (i) risk statistics; (ii) portfolio investability; and (iii) trading costs.

6.1. Risk statistics

Table 11, Panel A reports the major risk measures for the long-short value-weighted machine learning strategies. Their standard deviations of weekly returns are remarkably high, fitting into the range of 8.38% (FFNN) and 12.44% (RF). The associated Sharpe ratios are relatively high—their annualized values fall between 0.85 (PLS) and 1.66 (COMB). Notably, however impressive these values may appear, they fail to beat their counterparts in traditional asset classes—such as equities. For example, Gu et al. (2020) report annualized Sharpe ratios on long-short

¹⁰ An application of other models leads to consistent conclusions. The detailed results are available upon request.

Table 9

Interactions Between Machine Learning Predictions and Limits to Arbitrage.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
E(R)	0.897 (7.05)	0.390 (1.99)	0.307 (2.04)	0.376 (1.94)	0.348 (2.58)	0.442 (2.28)	0.557 (3.12)	0.439 (3.39)	0.426 (3.09)
beta						0.001 (0.22)	−0.001 (−0.27)	0.002 (1.16)	−0.001 (−0.24)
size						0.000 (0.39)	0.000 (1.03)	0.000 (1.46)	0.000 (0.64)
r22_1						0.004 (1.72)	0.003 (1.19)	0.000 (−0.15)	0.003 (1.26)
ivol _D		−0.009 (−2.06)				−0.011 (−2.57)			
E(R) × ivol _D		0.677 (2.86)				0.668 (2.84)			
bidask _D			−0.008 (−1.88)				−0.004 (−1.21)		
E(R) × bidask _D		0.741 (3.84)				0.559 (2.14)			
illiq _D				−0.009 (−2.05)				0.002 (0.50)	
E(R) × illiq _D				0.697 (2.97)				0.590 (2.58)	
lim _D					−0.004 (−0.89)				−0.005 (−1.07)
E(R) × lim _D					0.754 (3.54)				0.701 (3.35)
$\bar{R}^2$	1.85	2.52	2.28	2.51	2.43	3.31	3.48	3.56	3.22

The table reports the average slope coefficients from weekly predictive cross-sectional regressions following [Fama and MacBeth \(1973\)](#); where the dependent variable is a weekly cryptocurrency return, and the independent variables include expected returns $E(R)$ from the forecast combination (COMB) model, proxies of limits to arbitrage, and their interaction terms. The limits to arbitrage are represented by dummy variables that take a value of one if a given proxy is higher than a weekly median, or zero otherwise. The leftmost column indicates the dummies for four different proxies for limits to arbitrage: idiosyncratic volatility ($ivol_D$), illiquidity ($illiq_D$), bid-ask spread ($bidask_D$), and a composite measure (lim_D). Specifications (6) to (9) incorporate also control variables corresponding with the three-factor model of [Liu et al. \(2022\)](#): market beta (β), size ($size$), and momentum ($r22_1$). The numbers in parentheses are [Newey and West \(1987\)](#) adjusted t -statistics. The bottom row presents the average weekly cross-sectional adjusted R^2 coefficient ($\bar{R}^2$) expressed in percentage terms. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018.

machine learning strategies for the U.S. stocks of up to 2.45, and [Leippold et al. \(2022\)](#) record values of up to 3.45 for the Chinese market.

Notably, the long-short cryptocurrency strategies also exhibit asymmetrical risks—which are not fully captured by the standard deviation measure. For example, maximum weekly losses on all strategies are substantial; in many cases, an investor must be prepared to lose >30% in one week. The worst performer in this regard is FFNN, with a maximum weekly loss of 50.62%. This tail risk is also manifested in prolonged and deep maximum drawdowns. In most cases, the investors would have to survive a loss of >70–80% of the invested capital.¹¹ Hence, the potential tail risk in cryptocurrency machine-learning strategies may be substantial.

6.2. Long-short portfolio feasibility

[Table 11](#), Panel B focuses on the average cryptocurrency capitalization of different quintiles. The bottom quintile typically contains bigger coins, with an average market value typically between three and nine billion USD. Nevertheless, as seen in both [Table 4](#) and [Fig. 6](#), the abnormal returns concentrate mainly in the long leg, i.e., the one with the highest expected return. In this case, however, the capitalizations are noticeably smaller. The average cryptocurrency market value in the bottom quintile is roughly 100 to 1800 million USD.

[Table 11](#), Panel C offers an additional perspective on this issue by reporting the average proportion of the total market capitalization captured by different quintiles. The short side of the trade typically account for the biggest (or one of the biggest) chunk of the

cryptocurrency universe, representing up to 80.9%—for LASSO—of its aggregate market value. On the other hand, the bottom quintile—the primary source of alphas in the long-short strategies—accounts for between 0.7% (RF) and 15.4% (FFNN) of the market. Notably, this subset does not capture >3% of the aggregate cryptocurrency capitalization for half of the strategies.

To sum up, a practical implementation of long-short cryptocurrency strategies may pose substantial challenges. It hangs on taking massive long positions in very small coins that represent a minuscule fraction of the entire market. The feasibility of such operations may be questionable—or at least challenging.

6.3. Trading costs

Machine learning strategies typically require active trading and frequent portfolio reconstructions ([Gu et al., 2020](#); [Leippold et al., 2022](#)). This, in turn, may incur substantial trading costs—especially since the strategies derived their profitability primarily from small coins. High portfolio turnover and trading costs are the enemies of many quantitative strategies, impeding their performance and even calling into question their validity ([Novy-Marx & Velikov, 2016](#)). Does this problem haunt cryptocurrency strategies as well?

6.3.1. Portfolio turnover

[Table 11](#), Panel D demonstrates the portfolio turnover of quintile portfolios formed on machine learning forecasts. We calculate the turnover statistic as the average of portfolio share that needs to be replaced each week:

$$PT_t = \frac{1}{2} \sum_{i=1}^n |w_{i,t-1} \times (1 + r_{i,t}) - w_{i,t}|, \quad (9)$$

¹¹ Since the considered strategies represent long-short portfolios, the maximum drawdown losses are cumulated additively.

Table 10
Bivariate Portfolio Sorts on Limits to Arbitrage and Predicted Returns.

Panel A: Average returns														
	Equal-weighted portfolios							Value-weighted portfolios						
	Low E (R)	Mid E (R)	High E (R)	High-Low R	t-statR	High-Low α_3	t-stata	Low E (R)	Mid E (R)	High E (R)	High-Low R	t-stat	High-Low α_3	t-stata
<i>Panel A.1: Limits to arbitrage</i>														
Low <i>lim</i>	0.50	1.05	1.23	0.73	(2.06)	0.48	(1.50)	0.37	1.05	1.44	1.07	(2.26)	1.11	(2.24)
Medium <i>lim</i>	0.65	0.94	1.87	1.22	(3.13)	0.97	(2.81)	0.41	0.67	1.56	1.15	(1.70)	0.88	(1.30)
High <i>lim</i>	0.33	0.66	2.87	2.54	(5.08)	2.41	(4.40)	−0.05	0.91	3.53	3.58	(2.94)	3.31	(2.53)
High-Low <i>lim</i>				1.81	(2.89)	1.94	(3.06)				2.52	(1.87)	2.20	(1.56)
<i>Panel A.2: Illiquidity</i>														
Low <i>illiq</i>	0.07	0.73	1.36	1.30	(3.53)	1.08	(3.24)	−0.03	0.93	1.67	1.71	(3.06)	1.72	(2.95)
Medium <i>illiq</i>	0.57	0.95	1.59	1.02	(2.16)	0.70	(1.85)	0.95	0.81	1.18	0.23	(0.36)	−0.03	(−0.06)
High <i>illiq</i>	0.84	0.97	3.02	2.18	(4.62)	2.09	(3.74)	0.72	1.22	3.51	2.79	(2.65)	2.84	(2.25)
High-Low <i>illiq</i>				0.88	(1.49)	1.00	(1.47)				1.08	(0.89)	1.11	(0.76)
<i>Panel A.3: Bid-ask spread</i>														
Low <i>bidask</i>	0.80	1.02	1.41	0.61	(1.72)	0.42	(1.29)	0.58	0.83	1.24	0.66	(1.31)	0.66	(1.32)
Medium <i>bidask</i>	0.48	0.76	1.57	1.10	(2.86)	0.84	(2.17)	0.29	0.99	1.81	1.52	(2.99)	1.37	(2.08)
High <i>bidask</i>	0.21	0.88	2.99	2.78	(5.98)	2.60	(5.48)	−0.07	1.10	3.41	3.47	(2.97)	3.26	(2.80)
High-Low <i>bidask</i>				2.17	(3.75)	2.18	(3.80)				2.81	(2.21)	2.60	(2.02)
<i>Panel A.4: Idiosyncratic volatility</i>														
Low <i>ivol</i>	0.85	1.23	1.48	0.63	(1.62)	0.34	(1.08)	0.64	1.05	1.37	0.73	(1.78)	0.69	(1.69)
Medium <i>ivol</i>	0.76	0.91	1.35	0.59	(1.60)	0.38	(0.99)	0.45	1.01	1.56	1.11	(1.76)	1.09	(1.53)
High <i>ivol</i>	−0.13	0.51	3.13	3.26	(7.57)	3.14	(6.06)	−0.57	0.32	3.31	3.88	(3.20)	3.69	(3.21)
High-Low <i>ivol</i>				2.64	(4.52)	2.79	(4.51)				3.15	(2.37)	2.99	(2.44)

Panel B: Average cryptocurrency capitalization and market proportions							
	Average market cap (\$ bil)			Market proportion (%)			
	Low E(R)	Medium E(R)	High E(R)	Low E(R)	Medium E(R)	High E(R)	Sum
<i>Panel B.1: Limits to arbitrage</i>							
Low <i>lim</i>	7.25	9.53	3.27	28.55	40.42	13.91	82.88
Mid <i>lim</i>	0.27	0.29	0.14	7.13	5.88	2.29	15.30
High <i>lim</i>	0.21	0.11	0.06	0.92	0.70	0.20	1.82
<i>Panel B.2: Illiquidity</i>							
Low <i>illiq</i>	7.38	9.71	3.38	20.96	31.09	12.49	64.53
Mid <i>illiq</i>	0.26	0.14	0.06	14.39	15.08	3.85	33.33
High <i>illiq</i>	0.07	0.04	0.02	1.26	0.75	0.12	2.14
<i>Panel B.3: Bid-ask spread</i>							
Low <i>bidask</i>	7.55	9.81	3.41	34.04	45.02	15.44	94.49
Mid <i>bidask</i>	0.13	0.08	0.04	1.72	1.49	0.78	3.99
High <i>bidask</i>	0.06	0.03	0.02	0.83	0.48	0.20	1.52
<i>Panel B.4: Idiosyncratic volatility</i>							
Low <i>ivol</i>	4.39	6.54	2.63	5.19	5.29	2.97	13.45
Mid <i>ivol</i>	3.06	3.19	0.81	24.48	28.19	4.41	57.09
High <i>ivol</i>	0.27	0.16	0.03	6.97	13.44	9.06	29.46

The table presents the portfolios from bivariate sorts on different measures of limits to arbitrage and expected returns ($E(R)$) from the forecast combination (COMB) model. In the first step, we sort the cryptocurrencies on their limits to arbitrage using four different proxies: idiosyncratic volatility (*ivol*), illiquidity (*illiq*), bid-ask spread (*bidask*), and a composite measure (*lim*). Subsequently—within each of these subsets—we sort cryptocurrencies into Low $E(R)$, Medium $E(R)$, and High $E(R)$ terciles (as indicated in the top row) based on the COMB predictions. All portfolios are rebalanced weekly. Panel A displays the average weekly returns on the considered portfolios; it additionally presents the average returns ($H-L$ R) and three-factor model (Liu et al., 2022) alphas ($High-Low$ α_3) on the spread portfolio that buys (sells) the coins with the High (Low) $E(R)$. The left and right sections concern equal- and value-weighted strategies, respectively. The bottom rows of each panel present the differences in returns on the *High-Low* portfolios between the *High* and *Low* limits to arbitrage terciles. All returns and alphas are reported in percentages. The numbers in parentheses are Newey and West (1987) adjusted t -statistics. Panel B pertains to the relative importance of stocks in different portfolios. Its left side reports the average market value of coins in different categories (in \$ billion), and the right side illustrates the average proportion (in percentage) of the total market capitalization allocated therein. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018.

where $w_{i,t-1}$ and $w_{i,t}$ are the weights of cryptocurrency i in two consecutive weeks, and $r_{i,t}$ is the crypto return. We compute a one-sided (rather

than two-sided) turnover measure to avoid double-counting of buys and sells.¹²

Consistent with earlier machine learning studies, the portfolio

¹² Notably, the average weekly turnover computed this way for the case of long-short strategies may exceed 100% as the long and short legs' turnover are added up.

Table 11
Univariate Portfolio Sorts – Practical Perspective.

	OLS	PLS	LASSO	ENET	RF	GBRT	FFNN	COMB
<i>Panel A: Risk profile of the long-short strategies</i>								
Standard dev.(%)	10.01	8.54	11.74	12.00	12.44	14.11	8.38	10.56
Sharpe ratio	1.55	0.85	1.47	1.45	0.97	1.15	1.35	1.66
Max annual loss (%)	26.70	38.27	27.95	27.95	31.56	32.32	50.62	18.40
Max DD (%)	62.56	72.11	55.90	54.47	70.72	86.47	50.48	36.92
<i>Panel B: Average cryptocurrency capitalization (\$ billion)</i>								
Low	4.3	3.0	8.6	8.6	9.5	2.6	1.6	3.2
2	3.2	3.1	1.3	1.3	1.7	3.7	2.4	3.9
3	2.2	2.4	1.1	1.1	0.4	2.4	3.0	2.6
4	1.7	1.9	0.6	0.6	0.2	2.0	3.0	1.5
High	0.4	1.3	0.1	0.1	0.1	0.9	1.8	0.5
<i>Panel C: Market proportion (%)</i>								
Low	36.8	25.8	73.1	73.1	80.9	22.5	13.6	27.2
2	27.4	26.3	11.1	11.2	14.2	31.7	20.5	33.5
3	18.4	20.2	9.7	9.6	3.0	20.8	25.2	22.0
4	14.3	16.6	4.9	4.9	1.3	17.4	25.4	13.1
High	3.0	11.2	1.2	1.2	0.7	7.5	15.4	4.3
<i>Panel D: Portfolio turnover (%)</i>								
Low	36.3	36.0	24.7	24.7	54.3	36.9	87.2	44.9
2	67.8	62.3	49.1	48.9	71.1	57.1	91.5	69.8
3	69.8	67.9	52.5	52.3	79.5	56.0	92.5	73.4
4	66.0	64.4	50.2	50.1	77.4	53.3	91.8	71.6
High	48.7	50.4	38.4	38.3	73.8	37.7	86.6	54.3
High - Low	42.5	43.2	31.6	31.5	64.0	37.3	86.9	49.6

The table concerns practical aspects of the machine learning portfolio formed on cryptocurrency predictions. We sort the cryptocurrencies into quintiles based on predictions from eight models: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage and selection operator (LASSO), elastic net (ENET), gradient boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). *High (Low)* is the cryptocurrency quintile with the highest (lowest) return forecast, and *High-Low* is the long-short strategy that buys (sells) the top (bottom) quintile. All strategies are value-weighted and rebalanced weekly. Panel A reports different risks and performance metrics for weekly returns on the *High-Low* portfolios: standard deviation, annualized Sharpe ratio, maximum weekly loss, and maximum drawdown over the study period. Panel B displays the average market capitalization (in \$ billion) of cryptocurrencies in different quintiles, while Panel C shows their relative market proportion (in %). Finally, Panel D presents the average weekly portfolio turnover (in %)—interpreted as the share of the portfolio being replaced every week. The turnover for the long-short *High-Low* portfolio represents the average turnover in the long and short legs. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018.

turnover is substantial. The individual quintile portfolios typically require replacing about 40 to 70% of their value each week. As a result, the long-short strategies exhibit an average weekly turnover in each of the legs that falls between 31.5% (ENET) and 86.9% (FFNN). The approach based on the COMB model, previously frequently identified as a top performer, has a turnover ratio of 49.6%. Such a substantial portfolio rotation may negatively impact real-world portfolio performance.

6.3.2. Trading cost estimation

To quantify the transaction cost drag on the profitability of machine learning strategies, we continue with a more formal analysis. We calculate three different sensitivity measures to trading costs to ensure a comprehensive perspective. First, we compute the breakeven point for the trading costs. We estimate it as the average portfolio return divided by the average turnover.

Second, we turn to an estimation of net returns. In practice, obtaining detailed data on a large sample of cryptocurrency daily spreads and slippage rates may be challenging. Furthermore, the fixed transaction fees may vary across exchanges and investor types; they are also subject to various special offers. Hence, in the most simplistic approach, we follow Bianchi et al. (2022) and assume a fixed one-way fee of 30 (40) basis points for the long (short) leg as an all-inclusive approximation of transaction costs. Bianchi et al. (2022, p. 6) deem this measure “a set of fairly conservative transaction costs for a market maker.”

A vital shortcoming of the fixed trading costs is that they miss the inherent features of the machine learning strategies, which derive profits mainly from small and volatile cryptocurrencies. These coins are more likely to be illiquid and difficult to trade, so that the overall transaction costs may be underestimated. Hence, we also employ a third, more nuanced measure: the return net of variable transaction costs. In this framework, we dynamically estimate each cryptocurrency’s

effective spread using the methods of Corwin and Schultz (2012) and Abdi and Rinaldo (2017). This approach’s total one-way transaction cost is calculated as a sum of the bid-ask spread estimate (*bidask* defined as in Table 1), with an additional flat fee equal to 10 basis points. As seen from the properties of *bidask* in Table 2, this measure is much more conservative and assumes higher trading costs—especially for the tail of less liquid coins.

6.3.3. Cost-adjusted returns

Table 12 summarizes the role of trading costs in the long-short machine learning strategies illustrated using different approaches. Since the results for equal-weighted (Panel A) and value-weighted portfolios (Panel B) are relatively similar, let us focus on the value-weighted strategies. The breakeven trading costs range between 45 (FFNN) and 192 (ENET) basis points. These numbers are beyond the conservative trading costs assumptions in Bianchi et al. (2022); thus, they promise a potentially successful portfolio implementation.

Not surprisingly, all strategies produce positive and mostly significant mean returns when considering the fixed trading costs. The average profits range from 0.34% (FFNN) to 1.99% (ENET). More importantly, most strategies also survive the variable transaction cost approach. Admittedly, the drag on portfolio performance is substantial—the strategies lose about 57% of their gross profits on average. Furthermore, the mean returns on two long-short strategies—PLS and RF—are rendered insignificant, and the profits on the FFNN portfolio turn into losses. Nevertheless, most algorithms survive—continuing to produce sizeable and significant profits. The comprehensive COMB strategy generates an average weekly net return of 1.54% (*t*-stat = 2.55), i.e., 35% less when compared with the gross value. The top performers are the penalized regression; LASSO and ENET earn 2.04% (*t*-stat = 2.70) and 2.07% (*t*-stat = 2.67) per month, respectively.

To sum up, most of the cryptocurrency machine learning strategies

Table 12
Trading Costs of Machine Learning Strategies.

	OLS	PLS	LASSO	ENET	RF	GBRT	FFNN	COMB
<i>Panel A: Equal-weighted portfolios</i>								
Gross return (%)	2.35 (5.57)	1.29 (3.82)	1.94 (5.31)	1.92 (5.26)	1.52 (5.00)	1.88 (5.14)	0.88 (3.07)	1.87 (5.72)
Break-even TC (bps.)	156	89	179	178	126	141	36	105
Net return (flat TC)	1.82 (4.31)	0.78 (2.33)	1.56 (4.28)	1.54 (4.23)	1.09 (3.57)	1.41 (3.85)	0.02 (0.06)	1.25 (3.81)
Net return (variable TC)	1.72 (4.08)	0.62 (1.86)	1.62 (4.45)	1.61 (4.40)	1.13 (3.73)	1.25 (3.43)	−1.81 (−6.32)	0.96 (2.94)
<i>Panel B: Value-weighted portfolios</i>								
Gross return (%)	2.16 (3.04)	1.01 (1.91)	2.39 (3.16)	2.42 (3.12)	1.68 (2.39)	2.25 (2.61)	1.56 (3.41)	2.37 (3.93)
Break-even TC (bps.)	127	58	189	192	66	151	45	120
Net return (flat TC)	1.57 (2.22)	0.42 (0.79)	1.96 (2.60)	1.99 (2.57)	0.80 (1.14)	1.73 (2.01)	0.34 (0.75)	1.69 (2.80)
Net return (variable TC)	1.57 (2.21)	0.33 (0.62)	2.04 (2.70)	2.07 (2.67)	0.46 (0.65)	1.70 (1.97)	−1.59 (−3.47)	1.54 (2.55)

The table presents the impact of trading costs on the performance of long-short cryptocurrency machine-learning portfolios. We consider 10 machine learning models: ordinary least squares (OLS), partial least squares (PLS), the least absolute shrinkage and selection operator (LASSO), elastic net (ENET), gradient-boosted regression trees (GBRT), random forests (RF), feed-forward neural network (FFNN), and forecast combination (COMB). The strategies buy (sell) a quintile of cryptocurrencies with the highest (lowest) predicted return. All strategies are rebalanced weekly, and Panels A and B concern equal- and value-weighted portfolios—respectively. Gross return is the average weekly return on a long-short portfolio. Break-even TC (trading cost) is the level of one-way transaction costs at which the average returns decrease to zero. Net returns are estimated using two approaches: a) assuming a flat transaction fee that equals 30 (40) basis points for the long (short) positions (*Net return, flat TC*); and b) as a sum of half of the effective spread estimated using the methods of [Corwin and Schultz \(2012\)](#) and [Abdi and Rinaldo \(2017\)](#), and a flat fee of 10 basis points (*Net return, variable TC*). All the returns are reported in percentages, and the breakeven costs are expressed in basis points. The numbers in parentheses are *t*-statistics. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018.

seem robust to real-world impediments. Despite a high portfolio turnover, they endure even conservative trading costs. Nevertheless, this performance comes with two caveats. First, the strategies exhibit considerable tail risk and large drawdowns. Second, it requires taking substantial positions in the smallest cryptocurrencies—which pose practical difficulties. A successful practical implementation hinges on resolving these two challenges.

7. Concluding remarks

Both machine learning tools and cryptocurrency investments have attracted growing investor attention in recent years. Our study intersects these two fields. We combine asset pricing research with machine learning to gain new insights into return predictability in cryptocurrency markets. Using data ranging from 2017 to 2023, we calculate 40 cryptocurrency features. We use eight popular machine learning algorithms—including regularized regression, dimension reduction techniques, tree models, and neural networks. Using the cryptocurrency characteristics as inputs, we train the models to predict the cross-section of returns.

Our empirical analysis brings five main contributions. First, we demonstrate that machine learning models can be effectively applied to the cross-section of cryptocurrency returns. All our models' forecasts generate measurable economic gains and translate into profitable portfolios from univariate sorts. A value-weighted long-short quintile portfolio formed on the most comprehensive model—the forecast combination—generates a mean monthly return of 2.37%, which cannot be subsumed by popular factor models.

Second, we document remarkable differences between the return predictability in cryptocurrencies and traditional asset classes. First, the abnormal returns come almost solely from the long side of the trade buying the quintile of cryptocurrencies with the highest expected returns. Second, the profitability does not measurably decline over time. These two features place cryptocurrencies in stark contrast to stocks, where the alphas typically come from short legs of anomaly-based strategies and gradually shrink as markets become more efficient.

Third, we cast light on the crucial determinants of the cross-sectional variation in cryptocurrency returns. A variable importance analysis

allows us to pinpoint the crucial predictors. Most models derive their predictability from a few relatively simple features—such as market price, past alpha, momentum, and illiquidity. With respect to general categories of cryptocurrency features, what matters the most are past returns and liquidity.

Fourth, we provide insights into the sources of the cross-sectional return predictability in cryptocurrency markets. Our results align with the view that abnormal returns come from inefficiencies that cannot be easily arbitrated away. In line with this, they are concentrated in difficult-to-trade assets—where arbitrage constraints are more pronounced. High bid-ask spreads, illiquidity, and idiosyncratic risk visibly boost the return predictability.

Fifth, we assess the cryptocurrency machine learning strategies from a practical angle. Despite substantial portfolio turnover, we find them resilient to transaction costs—even when we assume relatively their relatively high levels. Nonetheless, the allegedly impressive performance has three hooks. First, the trading signals are short-lived. Consequently, reducing reconstruction frequency to two weeks markedly impedes their performance. Second, the strategies require taking considerable positions in the small and illiquid coins. Third, machine learning profits critically hinge on harvesting extreme returns on volatile cryptocurrencies. Reducing their role in a portfolio substantially decreases the returns. Resolving these three issues may be critical for the successful implementation of cryptocurrency machine-learning strategies.

Our study has direct implications for both individual investors and managers of investment funds, particularly hedge funds. Implementing the models proposed in the article paves the way for better portfolio performance. An analysis of the importance of variables shows which factors matter when constructing a portfolio consisting of cryptocurrencies. Finally, from a regulatory standpoint, the most important information is which factors are behind the success or failure of portfolio construction, as it allows for better alignment of regulatory frameworks. Our approach allows for a better understanding of the mechanisms that govern the cryptocurrency market.

Our research can be developed in several directions. The first avenue of exploration is to expand the set of machine learning models and explore the different approaches offered in the literature. An analysis

conducted on lower frequency data, such as monthly data, is also likely to provide new perspectives. However, this will only be possible in the distant future as we need more data to feed machine learning models. Any further research should take into account the postulate of explainable artificial intelligence.

Data availability

Additional data from the study is available at a dedicated website.

Appendix A. Appendix

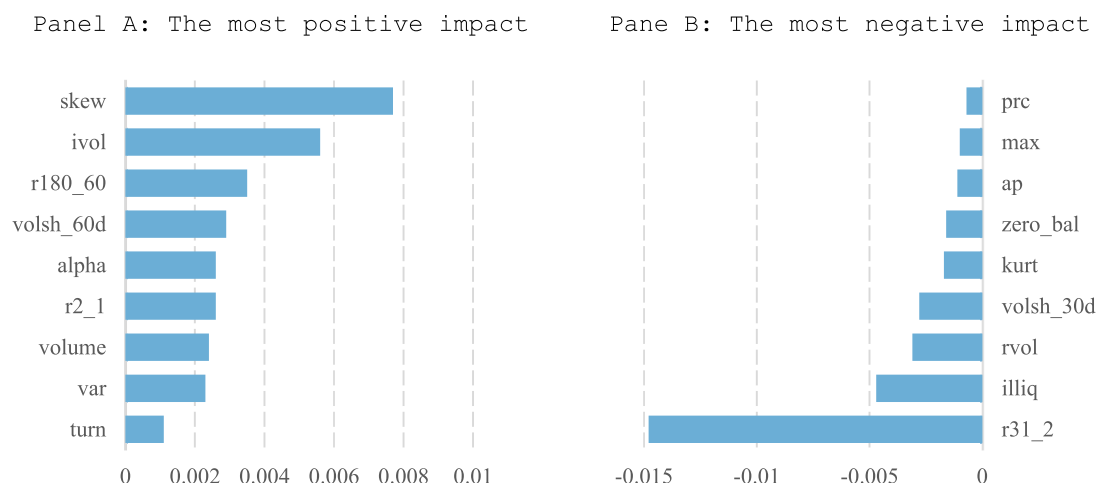


Fig. A1. Shapley Values for Model Variables.

The figure illustrates the variable importance measured with average weekly Shapley values the 10 input features with the most positive (Panel A) and negative (Panel B) impact on the overall forecasts. The predictions come from the ordinary least squares (OLS) model. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018.

Table A1
Statistical Properties of Factor Portfolios from Asset Pricing Models.

	Market factor (MKT)	Size (SIZE)	Momentum (MOM)
Mean	0.80	0.61	0.02
St. dev.	12.03	5.84	5.79
Skewness	-0.21	1.07	-0.11
Kurtosis	0.98	5.58	0.58
Minimum	-43.09	-15.47	-17.51
Maximum	38.93	35.87	16.69

The exhibit reports the basic statistical properties of the returns on factor portfolios from asset pricing models used to evaluate the machine learning strategies. The factor description is provided in Section 4.1, and the construction procedure follows Liu et al. (2022). The mean, standard deviation, minimum, and maximum weekly returns are expressed in percentage terms. The sample comprises 552 cryptocurrencies; the total study period is from July 1, 2017, to January 26, 2023, and the testing period starts on March 8, 2018.

References

- Abdi, F., & Rinaldo, A. (2017). A simple estimation of bid-ask spreads from daily close, high, and low prices. *Review of Financial Studies*, 30(12), 4437–4480.
- Akyildirim, E., Goncu, A., & Sensoy, A. (2021). Prediction of cryptocurrency returns using machine learning. *Annals of Operations Research*, 297, 3–36.
- Alessandretti, L., ElBahrawy, A., Aiello, L. M., & Baronchelli, A. (2018). Anticipating cryptocurrency prices using machine learning. *Complexity*, 1–16.
- Alexander, C., & Dakos, M. (2020). A critical investigation of cryptocurrency data and analysis. *Quantitative Finance*, 20(2), 173–188.
- Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56.
- Andersen, T. G., Bollerslev, T., & Meddahi, N. (2011). Realized volatility forecasting and market microstructure noise. *Journal of Econometrics*, 160(1), 220–234.
- Avramov, D., Cheng, S., & Metzker, L. (2023). Machine learning vs. economic restrictions: Evidence from stock return predictability. *Management Science*, 69(5), 2547–3155.
- Avramov, D., Chordia, T., Jostova, G., & Philipov, A. (2013). Anomalies and financial distress. *Journal of Financial Economics*, 108(1), 139–159.
- Azevedo, V., Kaiser, S., & Müller, S. (2023). Stock market anomalies and machine learning across the globe. *Journal of Asset Management*, 24, 419–441.
- Balcilar, M., Bouri, E., Gupta, R., & Roubaud, D. (2017). Can volume predict bitcoin returns and volatility? A quantiles-based approach. *Economic Modelling*, 64, 74–81.
- Bali, T. G., Cakici, N., & Whitelaw, R. F. (2011). Maxing out: Stocks as lotteries and the cross-section of expected returns. *Journal of Financial Economics*, 99(2), 427–446.
- Bali, T. G., Goyal, A., Huang, D., Jiang, F., & Wen, Q. (2022). Predicting corporate bond returns: Merton meets machine learning. In *Georgetown McDonough school of business research paper (3686164)* (pp. 20–110).
- Bartram, S. M., Djuranovic, L., Garratt, A., & Xu, Y. (2023). Mispricing and risk premia in currency markets. *Journal of Financial and Quantitative Analysis* (in press).

- Bates, J. M., & Granger, C. W. (1969). The combination of forecasts. *Journal of the Operational Research Society*, 20(4), 451–468.
- Bekiros, S., Gupta, R., & Majumdar, A. (2016). Incorporating economic policy uncertainty in US equity premium models: A nonlinear predictability analysis. *Finance Research Letters*, 18, 291–296.
- Bessembinder, H., Burt, A. P., & Hrdlicka, C. M. (). *Time series variation in the factor zoo*. Available at SSRN <https://ssrn.com/abstract=3992041>. or <https://doi.org/10.2139/ssrn.3992041>.
- Bianchi, D., & Babiak, M. (2022). A risk-based explanation of cryptocurrency returns. Available at SSRN 3935934.
- Bianchi, D., Babiak, M., & Dickerson, A. (2022). Trading volume and liquidity provision in cryptocurrency markets. *Journal of Banking & Finance*, 106547.
- Bianchi, D., Büchner, M., & Tamoni, A. (2021). Bond risk premiums with machine learning. *Review of Financial Studies*, 34(2), 1046–1089.
- Black, F. (1986). Noise. *Journal of Finance*, 41(3), 529–543.
- Borri, N. (2019). Conditional tail-risk in cryptocurrency markets. *Journal of Empirical Finance*, 50, 1–19.
- Borri, N., Massacci, D., Rubin, M., & Ruzzi, D. (2022). *Crypto risk premia*. Available at SSRN 4154627.
- Borri, N., & Shakhnov, K. (2022). The cross-section of cryptocurrency returns. *Review of Asset Pricing Studies*, 12(3), 667–705.
- Bouri, E., Kristoufek, L., Ahmad, T., & Shahzad, S. J. H. (2022). Microstructure noise and idiosyncratic volatility anomalies in cryptocurrencies. *Annals of Operations Research*, 1–27.
- Brauneis, A., Mestel, R., Riordan, R., & Theissen, E. (2021). How to measure the liquidity of cryptocurrency markets? *Journal of Banking & Finance*, 124, Article 106041.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45, 5–32.
- Burggraf, T., & Rudolf, M. (2021). Cryptocurrencies and the low volatility anomaly. *Finance Research Letters*, 40, Article 101683.
- Cai, C. X., & Zhao, R. (2024). Saliency theory and cryptocurrency returns. *Journal of Banking and Finance*, 159, Article 107052.
- Cakici, N., Fieberg, C., Metko, D., & Zaremba, A. (2023). Machine learning goes global: Cross-sectional return predictability in international stock markets. *Journal of Economic Dynamics and Control*, 155, Article 104725.
- Cakici, N., Fieberg, C., Metko, D., & Zaremba, A. (2024). Do anomalies really predict market returns? New data and new evidence. *Review of Finance*, 28(1), 1–44.
- Cakici, N., & Zaremba, A. (2022). Saliency theory and the cross-section of stock returns: International and further evidence. *Journal of Financial Economics*, 146(2), 689–725.
- Calluzzo, P., Moneta, F., & Topaloglu, S. (2019). When anomalies are publicized broadly, do institutions trade accordingly? *Management Science*, 65(10), 4555–4574.
- Chen, A. Y., & Velikov, M. (2023). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, 58(3), 968–1004.
- Chen, R., Lepori, G. M., Tai, C. C., & Sung, M. C. (2022). Can saliency theory explain investor behaviour? Real-world evidence from the cryptocurrency market. *International Review of Financial Analysis*, 84, Article 102419.
- Chesnay, F., & Jondeau, E. (2001). Does correlation between stock returns really increase during turbulent periods? *Economic Notes*, 30(1), 53–80.
- Chordia, T., Roll, R., & Subrahmanyam, A. (2008). Liquidity and market efficiency. *Journal of Financial Economics*, 87(2), 249–268.
- Chordia, T., Subrahmanyam, A., & Tong, Q. (2014). Have capital market anomalies attenuated in the recent era of high liquidity and trading activity? *Journal of Accounting and Economics*, 58(1), 41–58.
- Clemen, R. T. (1989). Combining forecasts: A review and annotated bibliography. *International Journal of Forecasting*, 5(4), 559–583.
- Cong, L. W., & He, Z. (2019). Blockchain disruption and smart contracts. *Review of Financial Studies*, 32(5), 1754–1797.
- Cong, L. W., Karolyi, G. A., Tang, K., & Zhao, W. (2022). *Value premium, network adoption, and factor pricing of crypto assets*. Available SSRN 3985631.
- Cont, R. (2010). Stylized properties of asset returns. In R. Cont (Ed.), *Encyclopedia of quantitative finance*.
- Coqueret, G. (2022). Persistence in factor-based supervised learning models. *Journal of Finance and Data Science*, 8, 12–34.
- Corwin, S. A., & Schultz, P. (2012). A simple way to estimate bid-ask spreads from daily high and low prices. *Journal of Finance*, 67(2), 719–760.
- Cosemans, M., & Frehen, R. (2021). Saliency theory and stock prices: Empirical evidence. *Journal of Financial Economics*, 140(2), 460–483.
- Datar, V. T., Naik, N. Y., & Radcliffe, R. (1998). Liquidity and stock returns: An alternative test. *Journal of Financial Markets*, 1(2), 203–219.
- Demir, E., Gozgor, G., Lau, C. K. M., & Vigne, S. A. (2018). Does economic policy uncertainty predict the bitcoin returns? An empirical investigation. *Finance Research Letters*, 26, 145–149.
- Deng, S., Huang, X., Zhu, Y., Su, Z., Fu, Z., & Shimada, T. (2023). Stock index direction forecasting using an explainable eXtreme gradient boosting and investor sentiments. *The North American Journal of Economics and Finance*, 64, Article 101848.
- Deng, S., Xiang, Y., Fu, Z., Wang, M., & Wang, Y. (2019). A hybrid method for crude oil price direction forecasting using multiple timeframes, dynamic time wrapping and genetic algorithm. *Applied Soft Computing*, 82, Article 105566.
- Diebold, F., & Mariano, R. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Diebold, F. X., & Shin, M. (2019). Machine learning for regularized survey forecast combination: Partially-egalitarian LASSO and its derivatives. *International Journal of Forecasting*, 35(4), 1679–1691.
- Dobrynskaya, V., & Dubrovskiy, M. (2022). *Cryptocurrencies meet equities: Risk factors and asset pricing relationships*. Higher School of Economics Research Paper No. WP BRP, 86.
- Dong, B., Jiang, L., Liu, J., & Zhu, Y. (2022). Liquidity in the cryptocurrency market and commonalities across anomalies. *International Review of Financial Analysis*, 81, Article 102097.
- Drobetz, W., & Otto, T. (2021). Empirical asset pricing via machine learning: Evidence from the European stock market. *Journal of Asset Management*, 22(7), 507–538.
- Fama, E. F., & French, K. R. (2008). Dissecting anomalies. *Journal of Finance*, 63(4), 1653–1678.
- Fama, E. F., & French, K. R. (2012). Size, value, and momentum in international stock returns. *Journal of Financial Economics*, 105(3), 457–472.
- Fama, E. F., & MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy*, 81(3), 607–636.
- Fan, G. F., Peng, L. L., & Hong, W. C. (2018). Short-term load forecasting based on phase space reconstruction algorithm and bi-square kernel regression model. *Applied Energy*, 224, 13–33.
- Fang, F., Chung, W., Ventre, C., Basios, M., Kanthan, L., Li, L., & Wu, F. (2021). Ascertaining price formation in cryptocurrency markets with machine learning. *European Journal of Finance*, 1–23.
- Fieberg, C., Günther, S., Poddig, T., & Zaremba, A. (2024). Non-standard errors in the cryptocurrency world. *International Review of Financial Analysis*, 92, Article 103106.
- Fieberg, C., Metko, D., Poddig, T., & Loy, T. (2023). Machine learning techniques for cross-sectional equity returns' prediction. *OR Spectrum*, 45, 289–323.
- Freyberger, J., Neuhierl, A., & Weber, M. (2020). Dissecting characteristics nonparametrically. *The Review of Financial Studies*, 33(5), 2326–2377.
- Garfinkel, J. A., Hsiao, L., & Hu, D. (2023). *Disagreement and the cross section of cryptocurrency returns*. Available at SSRN 4345640.
- George, T. J., & Hwang, C. Y. (2004). The 52-week high and momentum investing. *The Journal of Finance*, 59(5), 2145–2176.
- Giglio, S., Kelly, B., & Xiu, D. (2022). Factor models, machine learning, and asset pricing. *Annual Review of Financial Economics*, 14, 337–368.
- Grobys, K., & Junttila, J. (2021). Speculation and lottery-like demand in cryptocurrency markets. *Journal of International Financial Markets Institutions and Money*, 71, Article 101289.
- Grobys, K., & Sapkota, N. (2019). Cryptocurrencies and momentum. *Economics Letters*, 180, 6–10.
- Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.
- Gu, S., Kelly, B., & Xiu, D. (2021). Autoencoder asset pricing models. *Journal of Econometrics*, 222(1), 429–450.
- Gupta, R., Mwamba, J. W. M., & Wohar, M. E. (2018). The role of partisan conflict in forecasting the U.S. equity premium: A nonparametric approach. *Finance Research Letters*, 25, 131–136.
- Han, S. (2023). Is liquidity risk priced in cryptocurrency markets? *Applied Economics Letters*, 30(17), 2481–2487.
- Hanauer, M. X., & Kalsbach, T. (2023). Machine learning and the cross-section of emerging market stock returns. *Emerging Markets Review*, 55, Article 101022.
- Harvey, C. R., Liu, Y., & Zhu, H. (2016). ... and the cross-section of expected returns. *The Review of Financial Studies*, 29(1), 5–68.
- Hastie, T., Tibshirani, R., & Friedman, J. (2008). *The elements of statistical learning* (2nd ed.). New York: Springer.
- Hoang, L. T., & Baur, D. G. (2022). Loaded for bear: Bitcoin private wallets, exchange reserves and prices. *Journal of Banking & Finance*, 144, Article 106622.
- Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 55–67.
- Hong, H., Lim, T., & Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1), 265–295.
- Hou, K., Xue, C., & Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5), 2019–2133.
- Huang, J. Z., Huang, W., & Ni, J. (2019). Predicting bitcoin returns using high-dimensional technical indicators. *The Journal of Finance and Data Science*, 5(3), 140–155.
- Israel, R., & Moskowitz, T. J. (2013). The role of shorting, firm size, and time on market anomalies. *Journal of Financial Economics*, 108(2), 275–301.
- Jacobs, H. (2016). Market maturity and mispricing. *Journal of Financial Economics*, 122(2), 270–287.
- Jacobs, H., & Müller, S. (2020). Anomalies across the globe: Once public, no longer existent? *Journal of Financial Economics*, 135(1), 213–230.
- Jaeger, M., Krügel, S., Marinelli, D., Papenbrock, J., & Schwendner, P. (2021). Interpretable machine learning for diversified portfolio construction. *The Journal of Financial Data Science*, 3(3), 31–51.
- Jaquart, P., Dann, D., & Weinhardt, C. (2021). Short-term bitcoin market prediction via machine learning. *Journal of Finance and Data Science*, 7, 45–66.
- Jia, B., Goodell, J. W., & Shen, D. (2022). Momentum or reversal: Which is the appropriate third factor for cryptocurrencies? *Finance Research Letters*, 45, Article 102139.
- Jia, Y., Liu, Y., & Yan, S. (2021). Higher moments, extreme returns, and cross-section of cryptocurrency returns. *Finance Research Letters*, 39, Article 101536.
- Jones, C. S., & Mo, H. (2021). Out-of-sample performance of mutual fund predictors. *Review of Financial Studies*, 34(1), 149–193.
- Kelly, B., & Pruitt, S. (2013). Market expectations in the cross-section of present values. *Journal of Finance*, 68(5), 1721–1756.
- Kelly, B., & Pruitt, S. (2015). The three-pass regression filter: A new approach to forecasting using many predictors. *Journal of Econometrics*, 186(2), 294–316.
- Kelly, B. T., Malamud, S., & Zhou, K. (2024). The virtue of complexity in return prediction. *Journal of Finance*, 79(1), 459–503.
- Kelly, B. T., Pruitt, S., & Su, Y. (2019). Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3), 501–524.

- Keloharju, M., Linnainmaa, J. T., & Nyberg, P. (2016). Return seasonalities. *Journal of Finance*, 71(4), 1557–1590.
- Kingma, D. P., & Ba, J. (2014). *Adam: A method for stochastic optimization*. arXiv preprint arXiv:1412.6980.
- Lam, F. E. C., & Wei, K. J. (2011). Limits-to-arbitrage, investment frictions, and the asset growth anomaly. *Journal of Financial Economics*, 102(1), 127–149.
- Leippold, M., Wang, Q., & Zhou, W. (2022). Machine learning in the Chinese stock market. *Journal of Financial Economics*, 145(2A), 64–82.
- Leirvik, T. (2021). Cryptocurrency returns and the volatility of liquidity. *Finance Research Letters*, 44, Article 102031.
- Leitch, G., & Tanner, J. E. (1991). Economic forecast evaluation: profits versus the conventional error measures. *The American Economic Review*, 580–590.
- Lewellen, J., & Nagel, S. (2006). The conditional CAPM does not explain asset-pricing anomalies. *Journal of Financial Economics*, 82(2), 289–314.
- Li, Y., Urquhart, A., Wang, P., & Zhang, W. (2021). MAX momentum in cryptocurrency markets. *International Review of Financial Analysis*, 77, Article 101829.
- Liebi, L. J. (2022). Is there a value premium in cryptoasset markets? *Economic Modelling*, 109, Article 105777.
- Lin, C. H., Yen, K. C., & Cheng, H. P. (2021). Lottery-like momentum in the cryptocurrency market. *The North American Journal of Economics and Finance*, 58, Article 101552.
- Linnainmaa, J. T., & Roberts, M. R. (2018). The history of the cross-section of stock returns. *Review of Financial Studies*, 31(7), 2606–2649.
- Liu, W., Liang, X., & Cui, G. (2020). Common risk factors in the returns on cryptocurrencies. *Economic Modelling*, 86, 299–305.
- Liu, Y., Li, Z., Nekhili, R., & Sultan, J. (2023). Forecasting cryptocurrency returns with machine learning. *Research in International Business and Finance*, 64, Article 101905.
- Liu, Y., & Tsyvinski, A. (2021). Risks and returns of cryptocurrency. *Review of Financial Studies*, 34(6), 2689–2727.
- Liu, Y., Tsyvinski, A., & Wu, X. (2022). Common risk factors in cryptocurrency. *Journal of Finance*, 77(2), 1133–1177.
- Llorente, G., Michaely, R., Saar, G., & Wang, J. (2002). Dynamic volume-return relation of individual stocks. *Review of Financial Studies*, 15(4), 1005–1047.
- Lucchini, L., Alessandretti, L., Lepri, B., Gallo, A., & Baronchelli, A. (2020). From code to market: Network of developers and correlated returns of cryptocurrencies. *Science Advances*, 6(51), eabd2204.
- Lv, S. X., & Wang, L. (2023). Multivariate wind speed forecasting based on multi-objective feature selection approach and hybrid deep learning model. *Energy*, 263, Article 126100.
- McLean, R. D. (2010). Idiosyncratic risk, long-term reversal, and momentum. *Journal of Financial and Quantitative Analysis*, 45(4), 883–906.
- McLean, R. D., & Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.
- Menkveld, A. J., Dreber, A., Holzmeister, F., Huber, J., Johannesson, M., Kirchler, M., ... Weitzel, U. (2021). Non-standard errors. *Journal of Finance* (in press).
- Miller, M. H., & Scholes, M. S. (1982). Dividends and taxes: Some empirical evidence. *Journal of Political Economy*, 90(6), 1118–1141.
- Mohrschladt, H. (2021). The ordering of historical returns and the cross-section of subsequent returns. *Journal of Banking & Finance*, 125, Article 106064.
- Neely, C. J., Rapach, D. E., Tu, J., et al. (2014). Forecasting the equity risk premium: The role of technical indicators. *Management Science*, 60(7), 1772–1791.
- Newey, W. K., & West, K. D. (1987). A simple positive definite, heteroscedasticity, and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Novy-Marx, R., & Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1), 104–147.
- Ozdamar, M., Akdeniz, L., & Sensoy, A. (2021). Lottery-like preferences and the MAX effect in the cryptocurrency market. *Financial Innovation*, 7(1), 1–27.
- Pagnotta, E., & Buraschi, A. (2018). *An equilibrium valuation of bitcoin and decentralized network assets*. Available at SSRN 3142022.
- Petropoulos, F., Apiletti, D., Assimakopoulos, V., Babai, M. Z., Barrow, D. K., Taieb, S. B., ... Ziel, F. (2022). Forecasting: Theory and practice. *International Journal of Forecasting*, 38(3), 705–871.
- Pontiff, J. (2006). Costly arbitrage and the myth of idiosyncratic risk. *Journal of Accounting and Economics*, 42(1–2), 35–52.
- Rad, H., Low, R. K. Y., Miffre, J., & Faff, R. (2023). The commodity risk premium and neural networks. *Journal of Empirical Finance*, 74, Article 101433.
- Rudkin, S., Rudkin, W., & Dlotko, P. (2023). On the topology of cryptocurrency markets. *International Review of Financial Analysis*, 89, Article 102759.
- Sadka, R., & Scherbina, A. (2007). Analyst disagreement, mispricing, and liquidity. *Journal of Finance*, 62(5), 2367–2403.
- Schwert, G. W. (2003). Anomalies and market efficiency. *Handbook of the Economics of Finance*, 1, 939–974.
- Shen, D., Urquhart, A., & Wang, P. (2020). A three-factor pricing model for cryptocurrencies. *Finance Research Letters*, 34, Article 101248.
- Soehbag, A., van Vliet, B., & Vermijmeren, P. (2023). *Non-standard errors in asset pricing: Mind your sorts*. Available at SSRN 4136672.
- Stambaugh, R. F., Yu, J., & Yuan, Y. (2012). The short of it: Investor sentiment and anomalies. *Journal of Financial Economics*, 104(2), 288–302.
- Stambaugh, R. F., Yu, J., & Yuan, Y. (2015). Arbitrage asymmetry and the idiosyncratic volatility puzzle. *Journal of Finance*, 70(5), 1903–1948.
- Struck, C., & Cheng, E. (2020). The cross section of commodity returns: A nonparametric approach. *Journal of Financial Data Science*, 2(3), 86–103.
- Timmermann, A. (2006). Forecast combinations. *Handbook of Economic Forecasting*, 1, 135–196.
- Tobek, O., & Hronec, M. (2021). Does it pay to follow anomalies research? Machine learning approach with international evidence. *Journal of Financial Markets*, 56, Article 100588.
- Tzouvanas, P., Kizys, R., & Tsend-Ayush, B. (2020). Momentum trading in cryptocurrencies: Short-term returns and diversification benefits. *Economics Letters*, 191, Article 108728.
- Walter, D., Weber, R., & Weiss, P. (2022). *Non-standard errors in portfolio sorts*. Available at SSRN 4164117.
- Wang, Y., Wang, C., Sensoy, A., Yao, S., & Cheng, F. (2022). Can investors' informed trading predict cryptocurrency returns? Evidence from machine learning. *Research in International Business and Finance*, 62, Article 101683.
- Wu, J., Wang, J., Lu, H., Dong, Y., & Lu, X. (2013). Short-term load forecasting technique based on the seasonal exponential adjustment method and the regression model. *Energy Conversion and Management*, 70, 1–9.
- Yang, D., & Zhang, Q. (2000). Drift-independent volatility estimation based on high, low, open, and close prices. *Journal of Business*, 73(3), 477–492.
- Yun, K. K., Yoon, S. W., & Won, D. (2021). Prediction of stock price direction using a hybrid GA-XGBoost algorithm with a three-stage feature engineering process. *Expert Systems with Applications*, 186, Article 115716.
- Zaremba, A., Bilgin, M. H., Long, H., Mercik, A., & Szczygieski, J. J. (2021). Up or down? Short-term reversal, momentum, and liquidity effects in cryptocurrency markets. *International Review of Financial Analysis*, 78, Article 101908.
- Zaremba, A., Umutlu, M., & Maydybura, A. (2020). Where have the profits gone? Market efficiency and the disappearing equity anomalies in country and industry returns. *Journal of Banking & Finance*, 121, Article 105966.
- Zhang, W., & Li, Y. (2020). Is idiosyncratic volatility priced in cryptocurrency markets? *Research in International Business and Finance*, 54, Article 101252.
- Zhang, W., & Li, Y. (2023). Liquidity risk and expected cryptocurrency returns. *International Journal of Finance and Economics*, 28(1), 472–492.
- Zhang, W., Li, Y., Xiong, X., & Wang, P. (2021). Downside risk and the cross-section of cryptocurrency returns. *Journal of Banking & Finance*, 133, Article 106246.
- Zhou, X., Zhou, H., & Long, H. (2023). Forecasting the equity premium: Do deep neural network models work? *Modern Finance*, 1(1), 1–11.
- Zhu, B., Ye, S., Wang, P., Chevallier, J., & Wei, Y. M. (2022). Forecasting carbon price using a multi-objective least squares support vector machine with mixture kernels. *Journal of Forecasting*, 41(1), 100–117.
- Zou, H., & Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society, Series B: Statistical Methodology*, 67(2), 301–320.